Special DPP on Circle **DPP # 1**

1. Centres of the three circles

$x^2 + y^2 - 4x - 6y - 14 = 0$

$x^2 + y^2 + 2x + 4y - 5 = 0$

and $x^2 + y^2 - 10x - 16y + 7 = 0$:

(A) are the vertices of a right triangle (B) the vertices of an isosceles triangle which is not regular

(C) vertices of a regular triangle (D) are collinear

तीन वृतों

$x^2 + y^2 - 4x - 6y - 14 = 0$

$x^2 + y^2 + 2x + 4y - 5 = 0$

तथा $x^2 + y^2 - 10x - 16y + 7 = 0$ के केन्द्र होंगे :

(A) एक समकोण के शीर्ष (B) एक समद्विबाहु त्रिभुज के शीर्ष

(C) एक समबाहु त्रिभुज के शीर्ष (D) समरेखीय है

Ans. D

Sol. From the given equation, we get the centers A(2,3, B(–1,–2), C(5,8), respectively.

Using the distance formula as written below

$$d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

$$AB = \sqrt{(2+1)^2 + (3+2)^2}$$

$$AB = \sqrt{9 + 25}$$

$$AB = \sqrt{34} \text{ Units}$$

Similarly,

$$AC = \sqrt{(2-5)^2 + (3-8)^2}$$

$$AC = \sqrt{9 + 25}$$

$$AC = \sqrt{34} \text{ unit}$$

Similarly,

$$BC = \sqrt{(1-5)^2 + (-2-8)^2}$$

$BC = \sqrt{36+100}$

$BC = \sqrt{136} = 2\sqrt{34}$ unit

Since, $AB + AC = BC$

Hence, the points are collinear.

2. The equation of the image of the circle $x^2 + y^2 + 16x - 24y + 183 = 0$ by the line mirror $4x + 7y + 13 = 0$ is :
वृत्त $x^2 + y^2 + 16x - 24y + 183 = 0$ का रेखा $4x + 7y + 13 = 0$ में प्रतिबिम्ब है :
(A) $x^2 + y^2 + 32x - 4y + 235 = 0$ (B) $x^2 + y^2 + 32x + 4y - 235 = 0$
(C) $x^2 + y^2 + 32x - 4y - 235 = 0$ (D) $x^2 + y^2 + 32x + 4y + 235 = 0$

Ans. D

Sol. Find the image of the centre of the given circle $(-8, 12)$ in the line

$4x + 7y + 13 = 0$ which is $(-16, -2)$. Also radius $= \sqrt{64 + 144 - 183} = 5$

3. $2x^2 + 2y^2 + 2\lambda x + \lambda^2 = 0$ represents a circle for :
(A) each real value of λ (B) no real value of λ
(C) positive λ (D) negative λ

$2x^2 + 2y^2 + 2\lambda x + \lambda^2 = 0$ एक वृत है, तो :
(A) प्रत्येक λ के लिये (B) किसी वास्तविक λ के लिए नहीं
(C) धनात्मक λ के लिये (D) ऋणात्मक λ के लिये

Ans. A

Sol. (1) coefficient of x^2 = coefficient of y^2

$\Rightarrow 2 = 2$

(2) coefficient of $xy = 0$

Hence the given equation will represent a circle for all real values of λ.

4. Suppose that the equation of the circle having $(-3, 5)$ and $(5, -1)$ as end points of a diameter is $(x - a)^2 + (y - b)^2 = r^2$. Then $a + b + r$, $(r > 0)$ is :

माना एक वृत जिसके व्यास के छोर $(-3, 5)$ तथा $(5, -1)$ है का समीकरण $(x - a)^2 + (y - b)^2 = r^2$ है। तो $a + b + r$, $(r > 0)$ =
(A) 8 (B) 9 (C) 10 (D) 11

Ans. A

Sol. Centre $(1, 2)$; $r = \sqrt{16+9} = 5$; $a = 1$; $b = 2$; $r = 5 \Rightarrow a + b + r = 8$ Ans.

5. A square and an equilateral triangle have the same perimeter. Let A be the area of the circle circumscribed about the square and B be the area of the circle circumscribed about the triangle then the ratio $\frac{A}{B}$ is :

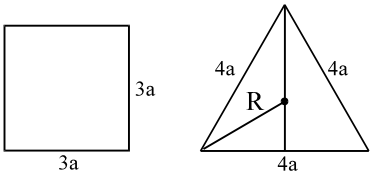
एक वर्ग तथा समबाहु त्रिभुज का परिमाप समान है। माना A वर्ग के परिवृत का क्षेत्रफल है तथा B त्रिभुज के परिवृत का क्षेत्रफल है, तो अनुपात $\frac{A}{B}$ है :

(A) $\frac{9}{16}$ (B) $\frac{3}{4}$ (C) $\frac{27}{32}$ (D) $\frac{3\sqrt{6}}{8}$

Ans. C

Sol. Radius of the circle circumscribing the square $= \frac{3\sqrt{2}\,a}{2}$

$A = \pi \cdot \frac{18a^2}{4} = \frac{9\pi a^2}{2}$



radius of the circle circumscribing triangle $= 2a \sec 30° = 2a \cdot \frac{2}{\sqrt{3}} = \frac{4a}{\sqrt{3}}$

$B = \pi \cdot \frac{16a^2}{3}$; Hence $\frac{A}{B} = \frac{9}{2} \cdot \frac{3}{16} = \frac{27}{32}$

6. The radii of the circles $x^2 + y^2 = 1$, $x^2 + y^2 - 2x - 6y = 6$ and $x^2 + y^2 - 4x - 12y = 9$ are in :

(A) A.P. (B) G.P. (C) H.P. (D) None

वृतो $x^2 + y^2 = 1$, $x^2 + y^2 - 2x - 6y = 6$ तथा $x^2 + y^2 - 4x - 12y = 9$ की त्रिज्याऐं :

(A) स.श्रे. में है (B) गु.श्रे. (C) ह.श्रे. (D) कोई नहीं

Ans. A

Sol. Given circles $x^2 + y^2 = 1$, $x^2 + y^2 - 2x - 6y = 6$ and $x^2 + y^2 - 4x - 12y = 9$

Let r_1, r_2 and r_3 be the radii o the circle (1), (2) and (3) respectively.

then $r_1 = 1$; $r_2 = \sqrt{(-1)^2 + (-3)^2 + 6} = 4$ and $r_3 = \sqrt{(-2)^2 + (-6)^2 + 9} = 7$

Clearly $r_2 - r_1 = 4 - 1 = 3 = r_3 - r_2$

Hence r_1, r_2, r_3 are in A.P.

7. A rhombus is inscribed in the region common to the two circles $x^2 + y^2 - 4x - 12 = 0$ and $x^2 + y^2 + 4x - 12 = 0$ with two of its vertices on the line joining the centres of the circles. The area of the rhombous is :

(A) $8\sqrt{3}$ sq.units (B) $4\sqrt{3}$ sq.units (C) $16\sqrt{3}$ sq.units (D) none

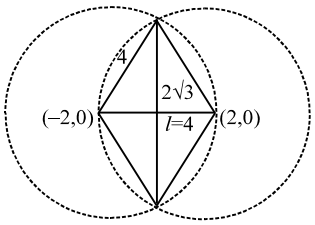
एक समचतुर्भुज दो वृतो $x^2 + y^2 - 4x - 12 = 0$ तथा $x^2 + y^2 + 4x - 12 = 0$ के मध्य उभयनिष्ठ क्षेत्र के अन्तः में स्थित है जहाँ इसके दो शीर्ष उभयनिष्ठ जीवा पर स्थित है। समचतुर्भुज का क्षेत्रफल हैं :

(A) $8\sqrt{3}$ sq.units (B) $4\sqrt{3}$ sq.units (C) $16\sqrt{3}$ sq.units (D) कोई नहीं

Ans. A

Sol. Circles with centre (2, 0) and (–2, 0) each with radius 4

$\Rightarrow$ y–axis is their common chord.



The inscribed rhombus has its diagonals equal to 4 and $4\sqrt{3}$

$\therefore \quad A = \frac{d_1 d_2}{2} = 8\sqrt{3}$

8. (6, 0), (0, 6) and (7, 7) are the vertices of a triangle. The circle inscribed in the triangle has the equation :

(6, 0), (0, 6) तथा (7, 7) त्रिभुज के शीर्ष है। उसके अन्तःखण्ड वृत का समीकरण है :

(A) $x^2 + y^2 - 9x + 9y + 36 = 0$ (B) $x^2 + y^2 - 9x - 9y + 36 = 0$

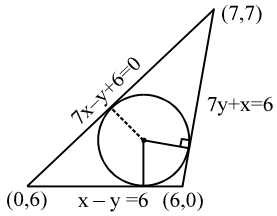
(C) $x^2 + y^2 + 9x - 9y + 36 = 0$ (D) $x^2 + y^2 - 9x - 9y - 36 = 0$

Ans. B

Sol. angle bisector of $\angle A$

angle bisector containing origin $8y - 6x - 12 = 0$

at $\angle A$ bisector not containing origin : $6x + 2y - 12 = 0$



find centre and radius

note that Δ is isosceles

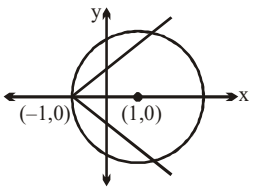
9. Number of points in which the graphs of $|y| = x + 1$ and $(x - 1)^2 + y^2 = 4$ intersect, is :

आरेखो $|y| = x + 1$ तथा $(x - 1)^2 + y^2 = 4$ के प्रतिच्छेदन बिन्दुओं की संख्या है :

(A) 1 (B) 2 (C) 3 (D) 4

Ans. C

Sol. The graph of $|y| = x + 1$ is two straight lines,
and the circle given by the other equation goes through their point of intersection.



10. The points (x_1, y_1), (x_2, y_2), (x_1, y_2) and (x_2, y_1) are always :
(A) collinear (B) concyclic (C) vertices of a square (D) vertices of a rhombus

बिन्दु (x_1, y_1), (x_2, y_2), (x_1, y_2) तथा (x_2, y_1) सदैव :
(A) समरेखीय (B) चक्रीय है (C) वर्ग के शीर्ष है (D) एक समचतुर्भुज के शीर्ष है

Ans. B

Sol. All the points lie on the circle $(x - x_1)(x - x_2) + (y - y_1)(y - y_2) = 0$

11. The line joining $(5, 0)$ to $(10\cos\theta, 10\sin\theta)$ is divided internally in the ratio $2 : 3$ at P. If θ varies then the locus of P is :
(A) a pair of straight lines (B) a circle
(C) a straight line (D) a second degree curve which is not a circle

$(5, 0)$ से $(10\cos\theta, 10\sin\theta)$ को मिलाने वाली रेखा को P $2 : 3$ में अन्तः विभाजन करता है। यदि θ चर है तो P का बिन्दुपथ है:
(A) रेखा युग्म (B) एक वृत
(C) सरल रेखा (D) एक दो कोटि का वक्र जो वृत नही है

Ans. B

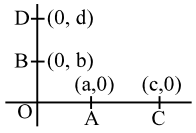
Sol. $x = 3 + 4\cos\theta$; $y = 4\sin\theta$

12. The points $A(a, 0)$, $B(0, b)$, $C(c, 0)$ and $D(0, d)$ are such that $ac = bd$ and a, b, c, d are all non-zero. Then the points :
(A) form a parallelogram (B) do not lie on a circle
(C) form a trapezium (D) are concyclic

बिन्दु $A(a, 0)$, $B(0, b)$, $C(c, 0)$ व $D(0, d)$ इस प्रकार है कि $ac = bd$ तथा a, b, c, d सभी अशून्य है, तो बिन्दु :
(A) एक समान्तर चतुर्भुज बनाते है (B) एक वृत पर नही है
(C) एक समलम्ब चतुर्भुज बनाते है (D) चक्रिय है

Ans. D

Sol. If $OA \cdot OC = OB \cdot OD$ (Power of point)
then points are concylic
$\therefore$ $a \cdot c = bd$ (true)



13. Four unit circles pass through the origin and have their centres on the coordinate axes. The area of the quadrilateral whose vertices are the points of intersection (in pairs) of the circles, is :

(A) 1 sq. unit (B) $2\sqrt{2}$ sq. units

(C) 4 sq. units (D) can not be uniquely determined, insufficient data

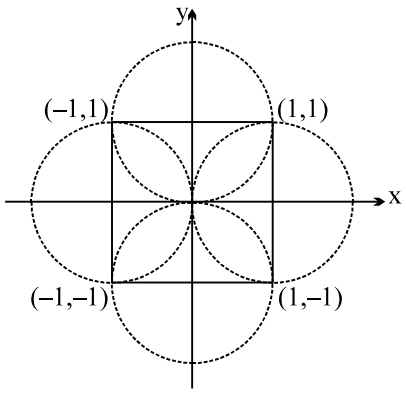
चार इकाई वृत मूल बिन्दु से गुजरते है तथा उसके केन्द्र अक्षों पर है। प्रतिच्छेदन बिन्दुओं द्वारा बनाये चतुर्भुज का क्षेत्रफल ज्ञात करेगें :

(A) 1 वर्ग इकाई (B) $2\sqrt{2}$ वर्ग इकाई

(C) 4 वर्ग इकाई (D) आंकड़े अप्रर्याप्त हैं

Ans. C

Sol. Ar. (square of sides 2) = 4 sq. units



14. Number of value(s) of A for which the system of equations $x^2 = y^2$ and $(x - A)^2 + y^2 = 1$ has exactly 3 solutions, is:

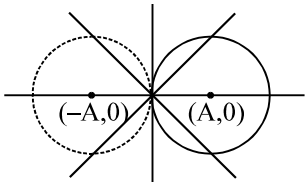
A के मानों की संख्या की समीकरणों $x^2 = y^2$ तथा $(x - A)^2 + y^2 = 1$ के ठीक 3 हल हो :

(A) 1 (B) 2 (C) 3 (D) 4

Ans. B

Sol. For exactly 3 solutions

$A = 1$ or $A = -1$



i.e. the circles touches the y axis with centre (1, 0) or (–1, 0) to cut the given pair at 3 points.

15. Coordinates of the centre of the circle which bisects the circumferences of the circles $x^2 + y^2 = 1$; $x^2 + y^2 + 2x - 3 = 0$ and $x^2 + y^2 + 2y - 3 = 0$ is :

उस वृत के केन्द्र के निर्देशांक जो वृतो $x^2 + y^2 = 1$; $x^2 + y^2 + 2x - 3 = 0$ तथा $x^2 + y^2 + 2y - 3 = 0$ की परिधियों को समभाजित करता है :

(A) $(-1, -1)$ (B) $(3, 3)$ (C) $(2, 2)$ (D) $(-2, -2)$

Ans. D

Sol. Let circle be $S \equiv x^2 + y^2 + 2gx + 2fy + c = 0$

$S_1 \equiv x^2 + y^2 = 1$

$S_2 \equiv x^2 + y^2 + 2x - 3 = 0$

$S_3 \equiv x^2 + y^2 + 2y - 3 = 0$

$\Rightarrow S - S_1 = 0$

is the equationn of chord of contact & it passes through centre of S_1

$\Rightarrow 2gx + 2fy + c + 1 = 0$

Satisfying $(0,0) \Rightarrow c = -1$,

Similarly $S - S_2 = 0$

$\Rightarrow (2g - 2)x + 2fy + 2 = 0$

Satisfying $(-1,0)$, we get $2 - 2g + 2 = 0$

$\Rightarrow g = 2$

Similarly, $S - S_3 = 0$

$\Rightarrow (2gx + (2f - 2)y + 2) = 0$

Satisfying $(0, -1)$, we get $\Rightarrow f = 2$

$\therefore$ Centre is $(-2, -2)$

16. Locus of all point P(x, y) satisfying $x^3 + y^3 + 3xy = 1$ consists of union of :

(A) a line and an isolated point (B) a line pair and an isolated point

(C) a line and a circle (D) a circle and an isolated point

बिन्दु P(x, y) का बिन्दुपथ जो $x^3 + y^3 + 3xy = 1$ तो संतुष्ट करता है, एक :

(A) एक रेखा तथा एक वियुक्त बिन्दु (B) एक रेखा युग्म तथा एक वियुक्त बिन्दु

(C) एक रेखा तथा एक वृत (D) एक वृत तथा एक वियुक्त बिन्दु

Ans. A

Sol. $x^3 + y^3 + (-1)^3 - 3(x)(y)(-1) = 0$

$(x + y - 1)[(x - y)^2 + (y + 1)^2 + (x + 1)^2] = 0$

$\Rightarrow$ $x + y = 1$ or $x = y = -1$

$\Rightarrow$ a line $x + y = 1$ or a point $(-1, -1)$ $\Rightarrow$ (A)

17. Find the coordinates of the centre and the radius of the circles whose equations are :

(a) $3x^2 + 3y^2 - 5x - 6y + 4 = 0$ (b) $4x^2 + 4y^2 - 16x - 12y + 21 = 0$.

निम्न समीकरण से वृत्त का केन्द्र व त्रिज्या ज्ञात कीजिये :

(a) $3x^2 + 3y^2 - 5x - 6y + 4 = 0$ (b) $4x^2 + 4y^2 - 16x - 12y + 21 = 0$.

Ans. (a) $\left(\frac{5}{6}, 1\right); \frac{1}{6}\sqrt{13}$ (b) $\left(2, \frac{3}{2}\right), 1$

Sol. (a) In $x^2 + y^2 + 2gx + 2fy + c = 0$

Centre is $(-g, -f)$ and radius = $\sqrt{g^2 + f^2 - c}$

Here equation is :

$$\Rightarrow x^2 + y^2 - \frac{5x}{3} - \frac{6y}{3} + \frac{4}{3} = 0$$

$$\therefore \text{ Centre} \rightarrow \left(\frac{5}{6}, \frac{6}{6}\right)$$

$$\rightarrow \left(\frac{5}{6}, 1\right)$$

$$\text{Radius} = \sqrt{\frac{25}{36} + 1 - \frac{4}{3}}$$

$$= \sqrt{\frac{25 + 36 - 48}{36}}$$

$$= \sqrt{\frac{13}{36}}$$

(b) Here equation is :

$$x^2 + y^2 - \frac{16x}{4} - \frac{12y}{4} + \frac{21}{4} = 0$$

$$x^2 + y^2 - 4x - 3y + \frac{21}{4} = 0$$

$$\therefore \text{ Centre} \rightarrow \left(2, \frac{3}{2}\right)$$

Radius $= \sqrt{4+\frac{9}{4}-\frac{21}{4}}$

$= \sqrt{4-\frac{12}{4}}$

$= 1$

18. If (4, 1) is an extremity of a diameter of the circle $x^2 + y^2 - 2x + 6y - 15 = 0$, find the co-ordinates of the other extremity of the diameter.

यदि (4, 1) वृत्त $x^2 + y^2 - 2x + 6y - 15 = 0$ के व्यास का एक सिरा हो, तो दूसरे सिरे के निर्देशांक ज्ञात कीजिये।

Ans. $(-2, -7)$

Sol. Centre of the circle is $(-g, -f)$

Centre is $C(1, -3)$

Let $A(x_1, y_1)$ be the other extremity.

Now, C is the midpoint between the two points $\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$

$\therefore (1,-3) = \left(\frac{x_1+4}{2}, \frac{y_1+1}{2}\right)$ .... [using mid-point formula]

Equating the x and y coordinates on both sides we get,

$1 = \frac{x_1+4}{2}$

$\Rightarrow 2 = x_1 + 4$

$\Rightarrow x_1 = -2$

$-3 = \frac{y_1+1}{2}$

$\Rightarrow -6 = y_1 + 1$

$\Rightarrow y_1 = -6 - 1 = -7$

$\therefore$ The other extremity of the diameter is $(-2, -7)$

19. Show that the point $(7, -5)$ lies on the circle $x^2 + y^2 - 6x + 4y - 12 = 0$ and find the co-ordinates of the other end of the diameter through this point.

सिद्ध कीजिए कि बिन्दु $(7, -5)$ वृत्त $x^2 + y^2 - 6x + 4y - 12 = 0$ पर स्थित है और इस बिन्दु से निकलने वाले वृत्त के व्यास के दूसरे सिरे के निर्देशांक ज्ञात कीजिए।

Ans. $(-1, 1)$

Sol. $\frac{7+\alpha}{2} = 3$

$\frac{-5+\beta}{2} = -2$

$(\alpha, \beta) = (-1, 1)$

20. Find the radius of the circle $(x\cos\alpha + y\sin\alpha - a)^2 + (x\sin\alpha - y\cos\alpha - b)^2 = k^2$ and if α varies, find the locus of its centre.

वृत्त $(x\cos\alpha + y\sin\alpha - a)^2 + (x\sin\alpha - y\cos\alpha - b)^2 = k^2$ की त्रिज्या ज्ञात कीजिए और यदि α चर है, तो केन्द्र का बिन्दुपथ ज्ञात कीजिए।

Ans. radius = k, Locus : $x^2 + y^2 = a^2 + b^2$

Sol. $x\cos^2\alpha + y^2\sin^2\alpha + a^2 + 2xy\sin\alpha\cos\alpha - 2ay\sin\alpha - 2ax\cos\alpha + x^2\sin^2\alpha + y^2\cos^2\alpha + b^2 - 2xy\sin\alpha\cos\alpha + 2by\cos\alpha - 2bx\sin\alpha = k^2$

$x^2 + y^2 - 2x(a\cos\alpha - b\sin\alpha) + 2y(b\cos\alpha - a\sin\alpha) + a^2 + b^2 - k^2 = 0$

$r = \sqrt{g^2 + f^2 - c}$

$r = \sqrt{(a\cos\alpha + b\sin\alpha)^2 + (b\cos\alpha - a\sin\alpha)^2 - a^2 - b^2 + k^2}$

$r = \sqrt{a^2 + b^2 + 0 - a^2 - b^2 + k^2}$

$r = k$

$x = a\cos\alpha + b\sin\alpha$

$y = -(a\sin\alpha - b\cos\alpha)$

$x^2 + y^2 = a^2\cos^2\alpha + 2ab\sin\alpha\cos\alpha + b^2\sin^2\alpha + a^2\sin^2\alpha - 2ab\sin\alpha\cos\alpha + b^2\cos^2\alpha$

$x^2 + y^2 = a^2 + b^2$

21. Find the equation to the circle which circumscribes the triangle formed by the lines $x = 3$; $x + y + 3 = 0$ and $x - y + 1 = 0$.

उस वृत का समीकरण ज्ञात करें जो रेखाओं $x = 3$; $x + y + 3 = 0$ तथा $x - y + 1 = 0$ द्वारा बनाए गए त्रिभुज का परिवृत है।

Ans. $x^2 + y^2 - 6x + 2y - 15 = 0$

Sol. The given equation of lines

$x + y = -3$ ...(i)

$x - y =$ ...(ii)

$x = 3$ ...(iii)

Let A, B, C are the point of intersection of lines (i) and (ii), (ii) and (iii) and (i) respectively

$\therefore$ A = (–2, –1), B(3,4) and C = (3, –6)

Now A circle $x^2 + y^2 + 2gx + 2fy + c = 0$ ...(A)

circumscribing the ΔABC

$\therefore 4 + 1 - 4g - 2f + c = 0$ ...(iv)

$9 + 16 + 6g + 8f + c = 0$ ...(v)

$9 + 36 + 6g - 12f + c = 0$ ...(vi)

Solving (iv), (v) and (vi), we get,

$g = -3$, $f = 1$ and $c = -15$

from (A)

The equation of required circle is $x^2 + y^2 - 6x + 2y - 15 = 0$

1. $y-1=m_1(x-3)$ and $y-3=m_2(x-1)$ are two family of straight lines, at right angles to each other. The locus of their point of intersection is :

$y-1=m_1(x-3)$ तथा $y-3=m_2(x-1)$ दो रेखा समुह है तो एक दूसरे के लम्बवत् है। उनके प्रतिच्छेदन बिन्दु का बिन्दुपथ है :

(A) $x^2+y^2-2x-6y+10=0$ (B) $x^2+y^2-4x-4y+6=0$

(C) $x^2+y^2-2x-6y+6=0$ (D) $x^2+y^2-4x-4y-6=0$

Ans. B

Sol. Given $y-1=m_1(x-3)$ and $y-3=m_2(x-1)$

$$m_1=\frac{y-1}{x-3}, m_2=\frac{y-3}{x-1}$$

As the lines are at right angled to each other

Slopes product $=-1$

Hence,

$m_1\times m_2=-1$

$$\Rightarrow \frac{y-1}{x-3}\times\frac{y-3}{x-1}=-1$$

$$\Rightarrow y^2-4y+3=-\left(x^2-4x+3\right)$$

$$\Rightarrow x^2+y^2-4x-4y+6=0$$

$\therefore$ the locus of intersection

$x^2+y^2-4x-4y+6=0$

2. The circle described on the line joining the points $(0,1)$, (a,b) as diameter cuts the x–axis in points whose abscissae are roots of the equation :

बिन्दुओं $(0,1)$, (a,b) को व्यास मानकर बनाये वृत का समीकरण x–अक्ष को वहाँ प्रतिच्छेद करेगा जिनके भुज निम्न समीकरण के मूल है :

(A) $x^2+ax+b=0$ (B) $x^2-ax+b=0$ (C) $x^2+ax-b=0$ (D) $x^2-ax-b=0$

Ans. B

Sol. The circle equation is given by

$(x - 0)(x - a) + (y - 1)(y - b) = 0$

$\Rightarrow x^2 + y^2 - y(b + 1) - ax + b = 0$

This equation cuts the x-axis at $y = 0$

$x^2 - ax + b = 0$

x is given by the above equation

The abscissa re roots of the above equation.

3. B and C are fixed points having co–ordinates (3, 0) and (– 3, 0) respectively . If the vertical angle BAC is 90º, then the locus of the centroid of the ΔABC has the equation :

B तथा C स्थिर बिन्दु हैं जिनके निर्देशांक (3, 0) तथा (– 3, 0) क्रमशः है। यदि BAC = 90º तो Δ ABC के केन्द्रक का बिन्दुपथ है :

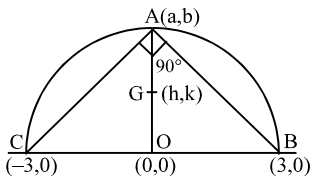
(A) $x^2 + y^2 = 1$ (B) $x^2 + y^2 = 2$ (C) $9(x^2 + y^2) = 1$ (D) $9(x^2 + y^2) = 4$

Ans. A

Sol. Let A (a, b) and G (h. k) Now A, G, O are collinear

$\Rightarrow h = \frac{2.0 + a}{3} \Rightarrow a = 3h$ and similarly $b = 3k$.

Now (a, b) lies on the circle $x^2 + y^2 = 9 \Rightarrow$ A



4. The ends of a quadrant of a circle have the coordinates (1, 3) and (3, 1) then the centre of the such a circle is :

एक वृत के एक चतुर्थांश के शीर्ष (1, 3) तथा (3, 1) है तो ऐसे वृत का केन्द्र है :

(A) (1, 1) (B) (2, 2) (C) (2, 6) (D) (4, 4)

Ans. A

Sol. Slope of AB = – 1; OM = 1

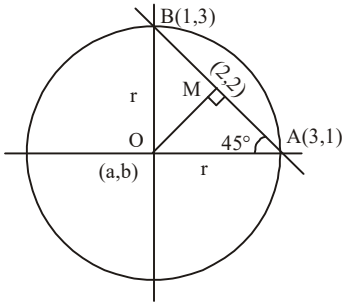
also $2r^2 = 4 + 4 = 8 \Rightarrow r = 2 \Rightarrow AM = \sqrt{2}$

$(AM)^2 + (OM)^2 = (OA)^2$

$2 + (a - 2)^2 + (b - 2)^2 = (a - 3)^2 + (b - 1)^2$

$2 - 4a - 4b + 8 = - 6a - 2b + 10$

$\Rightarrow a = b$



Also $(OA)^2 + (OB)^2 = (AB)^2$

$2[(a - 1)^2 + (a - 3)^2] = 8$

$\Rightarrow a = 1$ or $a = 3$

5. In the xy plane, the segment with end points (3, 8) and (–5, 2) is the diameter of the circle. The point (k, 10) lies on

the circle for :

(A) no value of k (B) exactly one integral k

(C) exacly one non integral k (D) two real values of k

xy तल में रेखाखण्ड (3, 8) तथा (–5, 2) वृत का व्यास है। बिन्दु (k, 10) वृत के अन्दर है, तो :

(A) किसी k के लिये नही (B) केवल एक पूर्णांक k के लिये

(C) केवल एक अपूर्णांक k के लिये (D) दो वास्तविक k के लिये

Ans. B

Sol. $(x - 3)(x + 5) + (y - 8)(y - 2) = 0$

$\Rightarrow x^2 + y^2 + 2x - 10y - 15 + 16 = 0$

$\Rightarrow x^2 + y^2 + 2x - 10y + 1 = 0$

Put (k, 10)

$k^2 + 2k + 1 = 0$

$k = -1$

6. A circle of radius 5 has its centre on the negative x-axis and passes through the point (2, 3). The intercept made by the circle on the y-axis is :

(A) 10 (B) $2\sqrt{21}$ (C) $2\sqrt{11}$ (D) imaginary y-intercept

(2, 3) से जाने वाले तथा 5 त्रिज्या वाले वृत का केन्द्र ऋणात्मक x अक्ष पर है तो वृत द्वारा y अक्ष पर काटा गये अन्तःखण्ड की लम्बाई होगी:

(A) 10 (B) $2\sqrt{21}$ (C) $2\sqrt{11}$ (D) काल्पनिक y-अन्तःखण्ड

Ans. B

Sol. Centre $(-a, 0)$, $a > 0$

$(2 + a)^2 + 9 = 25 \Rightarrow a = 2$

Equation of circle is $x^2 + y^2 + 4x - 21 = 0$

y-intercept $= 2\sqrt{f^2 - c} = 2\sqrt{21}$

7. The equation of a line inclined at an angle $\frac{\pi}{4}$ to the axis X, such that the two circles $x^2 + y^2 = 4$, $x^2 + y^2 - 10x - 14y + 65 = 0$ intercept equal lengths on it, is :

x अक्ष से $\pi/4$ पर झुकी हुई रेखा का समीकरण कि वह वृतो $x^2 + y^2 = 4$, $x^2 + y^2 - 10x - 14y + 65 = 0$ पर बराबर लम्बाई का अन्तःखण्ड काटती है :

(A) $2x - 2y - 3 = 0$ (B) $2x - 2y + 3 = 0$ (C) $x - y + 6 = 0$ (D) $x - y - 6 = 0$

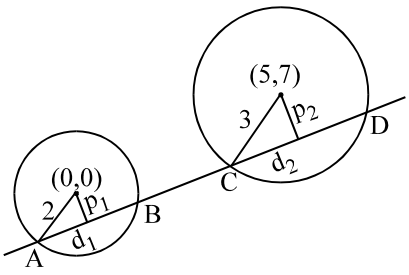
Ans. A

Sol. Let equation of line be $y = x + c$

$y - x = c$ ....(1)

perpendicular from (0, 0) on (1) is $\left|\frac{-c}{\sqrt{2}}\right| = \frac{c}{\sqrt{2}}$

In ΔAON, $\sqrt{2^2 - \left(\frac{c}{\sqrt{2}}\right)^2} = AN$



and in ΔCPM, $\sqrt{3^2 - 2 - \frac{c}{\sqrt{2}}} = CM$

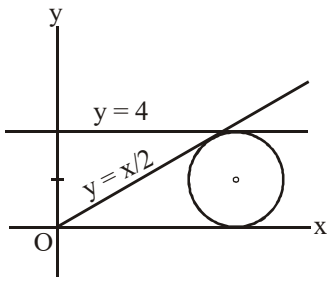
perpendicular from (5, 7) on line $y - x = c = \frac{2 - c}{\sqrt{2}}$

Given $AN = CM = 4 - \frac{c^2}{2} = 9 - \frac{(2-c)^2}{2} \Rightarrow c = -\frac{3}{2}$

$\therefore$ equation of line $y = x - \frac{3}{2}$ of $2x - 2y - 3 = 0$

8. The x-coordinate of the center of the circle in the first quadrant (see figure) tangent to the lines $y = \frac{1}{2}x$, $y = 4$ and the x-axis is :

चित्र में वृत के प्रथम चतुर्थांश में केन्द्र का x-निर्देशांक है जो रेखाओं $y = \frac{1}{2}x$, $y = 4$ तथा x-अक्ष को स्पर्श करता है :



(A) $4 + 2\sqrt{5}$ (B) $4 + \frac{8\sqrt{5}}{5}$ (C) $2 + \frac{6\sqrt{5}}{5}$ (D) $8 + 2\sqrt{5}$

Ans. A

Sol. $x - 2y = 0$ let centre is (k, 2)

$$\frac{|k-4|}{\sqrt{5}} = 2$$

$$|k - 4| = 2\sqrt{5}$$

$k = 4 + 2\sqrt{5} \Rightarrow$ A $\quad k = 4 - 2\sqrt{5}$ rejected

9. A(1, 0) and B(0, 1) and two fixed points on the circle $x^2 + y^2 = 1$. C is a variable point on this circle. As C moves, the locus of the orthocentre of the triangle ABC is :

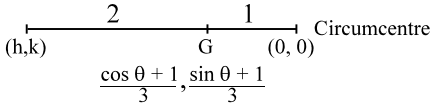
A(1, 0) तथा B(0, 1) दो नियत बिन्दु वृत $x^2 + y^2 = 1$ पर है। C एक चर बिन्दु वृत पर है। त्रिभुज ABC के लम्ब केन्द्र का बिन्दुपथ है :

(A) $x^2 + y^2 - 2x - 2y + 1 = 0$ (B) $x^2 + y^2 - x - y = 0$

(C) $x^2 + y^2 = 4$ (D) $x^2 + y^2 + 2x - 2y + 1 = 0$

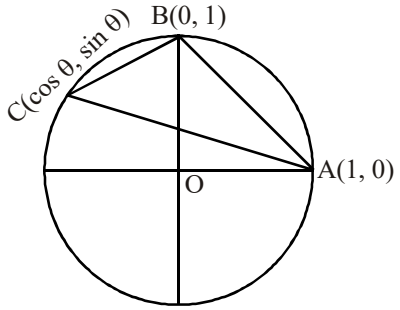
Ans. A

Sol. Let C $(\cos\theta, \sin\theta)$; H(h, k) is the orthocentre of the Δ ABC



$h = 1 + \cos\theta$

$k = 1 + \sin\theta$



$(x - 1)^2 + (y - 1)^2 = 1$

$x^2 + y^2 - 2x - 2y + 1 = 0$

10. $\frac{x - x_1}{\cos\theta} = \frac{y - y_1}{\sin\theta} = r$, represents :

(A) equation of a straight line, if θ is constant and r is variable

(B) equation of a circle, if r is constant and θ is a variable

(C) a straight line passing through a fixed point and having a known slope

(D) a circle with a known centre and a given radius

$\frac{x - x_1}{\cos\theta} = \frac{y - y_1}{\sin\theta} = r$ है :

(A) एक रेखा का समीकरण यदि θ नियतांक है एवं r चर है

(B) एक वृत का समीकरण यदि r नियतांक है एवं θ चर है

(C) एक रेखा का समीकरण है जो एक नियत बिन्दु से होकर गुजरती है एवं जिसकी ढाल ज्ञात है

(D) एक वृत का समीकरण है जिसका केन्द्र एवं त्रिज्या दोनों ज्ञात है

Ans. ABCD

Sol. We have given

$$\frac{x - x_1}{\cos\theta} = \frac{y - y_1}{\sin\theta} = r \qquad ...(1)$$

Case-I :

Consider,

$$\frac{x - x_1}{\cos\theta} = \frac{y - y_1}{\sin\theta} = r$$

If θ is constant and r is variable, then it represents an equation of straight line having equation

$$y - y_1 = \frac{\sin\theta}{\cos\theta}(x - x_1)$$

$$\Rightarrow y - y_1 = \tan\theta(x - x_1)$$

Hence, equation (1) is equation of straight line, if θ is constant and r is variable.

Also, a straight line passing through a fixed point and having a known slope.

Case-II :

Consider, $\frac{x - x_1}{\cos\theta} = \frac{y - y_1}{\sin\theta} = r$ represents the equation of circle in the polar form having equation.

$$(x - x_1)^2 + (y - y_1)^2 = r^2$$

which has coordinates of centre (x_1, y_1) and has radius r.

$\Rightarrow$ equation (1) represents equation of circle, if r is constant and θ is variable having centre as (x_1, y_1) and radius r.
So, from case I and II.

Option A, B, C and D are correct answers.

11. Let C be a circle $x^2 + y^2 = 1$. The line *l* intersects C at the point (–1, 0) and the point P. Suppose that the slope of

the line l is a rational number m. Number of choices for m for which both the coordinates of P are rational, is :

(A) 3 (B) 4 (C) 5 (D) infinitely many

माना C एक वृत $x^2 + y^2 = 1$ है। रेखा l वृत C को बिन्दु (–1, 0) एवं बिन्दु P पर प्रतिच्छेद करती है। माना रेखा l की परिमेय प्रवणता m है। m के लिये सम्भावित मानो की संख्या जिसके लिये बिन्दु P परिमेय है।

(A) 3 (B) 4 (C) 5 (D) अनन्त

Ans. D

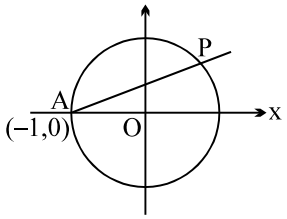
Sol. Equation of the line l is

$y - 0 = m(x + 1)$ ....(1)

solving it with $x^2 + y^2 = 1$

$x^2 + m^2(x + 1)^2 = 1$

$(m^2 + 1)x^2 + 2m^2x + (m^2 - 1) = 0,\ m \in Q$



$$x = \frac{-2m^2 \pm \sqrt{4m^4 - 4(m^4 - 1)}}{2(m^2 + 1)} = \frac{-2m^2 \pm 2}{2(m^2 + 1)}$$

taking – ve sign $x = -1$ (corresponding to A)

with + ve sign $x = \dfrac{1 - m^2}{1 + m^2}$

since $m \in Q$ hence x will be rational.

If x is rational then y is also rational from (1)

12. Consider the points P (2, 1) ; Q (0, 0) ; R (4, –3) and the circle $S : x^2 + y^2 - 5x + 2y - 5 = 0$

(A) exactly one point lies outside S (B) exactly two points lie outside S

(C) all the three points lie outside S (D) none of the point lies outside S

माना बिन्दु P (2, 1) ; Q (0, 0) ; R (4, –3) तथा वृत $S : x^2 + y^2 - 5x + 2y - 5 = 0$

(A) केवल एक बिन्दु S के बाह्य है (B) केवल दो बिन्दु S के बाह्य है

(C) सभी तीन बिन्दु S के बाह्य है (D) कोई बिन्दु S के बाहर नही है

Ans. D

Sol. Given points P (2, 1) ; Q (0, 0) ; R (4, –3)

$S : x^2 + y^2 - 5x + 2y - 5 = 0$

$$C = \left(\frac{5}{2}, -1\right)$$

distance $PC = \sqrt{\left(\frac{5}{2} - 2\right)^2 + (-1 - 1)^2}$ $r = \sqrt{\left(\frac{5}{2}\right)^2 + 1 + 5} = \sqrt{\frac{25}{4} + 1 + 5} = \sqrt{\frac{49}{4}} = \frac{7}{2}$

$$= \sqrt{\frac{1}{4}+4} = \frac{\sqrt{17}}{2} < r$$

distance CA $= \sqrt{\left(\frac{5}{2}\right)^2 + (1)^2} = \sqrt{\frac{25}{4}+1} = \frac{\sqrt{29}}{2} < r$

distance RC $= \sqrt{\left(4-\frac{5}{2}\right)^2 + (-3+1)^2} = \sqrt{\frac{9}{4}+4} = \frac{\sqrt{25}}{2} < r$

$\therefore$ all points line inside circle

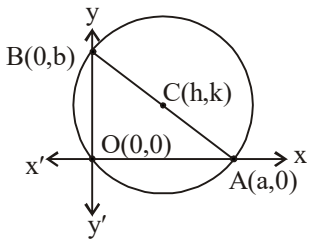
13. Find the equation to the circle which goes through the origin and cuts off intercepts equal to h and k from the positive parts of the axes.

उस वृत का समीकरण ज्ञात करें जो मूल बिन्दु से होकर गुजरे तथा जिसके द्वारा धनात्मक अक्षो पर काटे गये अन्तः खण्डो की लम्बाई h तथा k है।

Ans. $x^2 + y^2 - hx - ky = 0$

Sol. Since the circle is passes through origin and cut intercept a and b on x-axis and y-axis

$\Rightarrow$ Circle passes through the points A(a,0) and B(0,b).



Since, x-intercept is a and y-intercept is band the circle is passing through origin.

i.e., AB is the diameter of circle.

If(x_1, y_1) and (x_2,y_2) are end points of diameter of the circle then, equation of the circle is

$(x - x_1)(x - x_2) + (y - y_1)(y - y_2) = 0$

Here, A(a,0) and B(0,b) are the end points of diameter of the circle.

So, the equation of the circle is,

$(x - a)(x - 0) + (y - 0)(y - b) = 0$

$\Rightarrow (x - a)x + (y - b)y = 0$

$\Rightarrow x^2 + y^2 - ax - by = 0 - (1)$

Since, the intercepts a and b may negative.

$\Rightarrow$ Required equation of the circle is,

$x^2 + y^2 \pm ax \pm by = 0$

14. Find the equation to the circle which touches each positive axes at a distance 5 from the origin.
उस वृत का समीकरण ज्ञात करें जो धनात्मक अक्षो को मूल बिन्दु से 5 इकाई की दूरी पर स्पर्श करता है।

Ans. $x^2 + y^2 - 10x - 10y + 25 = 0$

Sol.

15. Find the equation to the circle which touches the axis of x and passes through the two points (1, – 2) and (3, – 4).

उस वृत का समीकरण ज्ञात करें जो x अक्ष को स्पर्श करें, एवं बिन्दुओं (1, – 2) तथा (3, – 4) से गुजरे।

Ans. $x^2 + y^2 - 6x + 4y + 9 = 0$, or $x^2 + y^2 + 10x + 20y + 25 = 0$

Sol. For the circle

$x^2 + y^2 + 2gx + 2fy + c = 0$

From the figure $f = r = \sqrt{g^2 + f^2 - c}$

$\Rightarrow g^2 = c$

And point (1, –2), (3, –4) are on the circle.

Then, we have

$5 + 2g - 4f + g^2 = 0$ ...(1)

$25 + 6g - 8f + g^2 = 0$ ...(2)

from (1) and (2)

$g^2 - 2g - 15 = 0$

$g = -5, g = 3$

So, $g = 3, f = 2, c = 9$

Equation of circle becomes $x^2 + y^2 - 6x + 4y + 9 = 0$

16. Find the equation to the circle which touches the axis of :
(a) x at a distance + 3 from the origin and intercepts a distance 6 on the axis of y.
(b) y at a distance – 3 from the origin and intercepts a length 8 on the axis of x.
(c) x, pass through the point (1, 1) and have line $x + y = 3$ as diameter.

उस वृत का समीकरण ज्ञात करें :

(a) जो x अक्ष को मूलबिन्दु से + 3 दूरी पर स्पर्श करे तथा y अक्ष पर 6 लम्बाई का अन्तःखण्ड काटे

(b) जो y अक्ष को मूलबिन्दु से – 3 दूरी पर स्पर्श करे तथा x अक्ष पर 8 लम्बाई का अन्तःखण्ड काटे

(c) जो x अक्ष को स्पर्श करे, बिन्दु (1, 1) से गुजरे तथा $x + y = 3$ उसका व्यास हो

Ans. (a) $x^2 + y^2 - 6x \pm 6\sqrt{2}\, y + 9 = 0$ (b) $x^2 + y^2 \pm 10x + 6y + 9 = 0$

(c) $x^2 + y^2 + 4x - 10y + 4 = 0$; $x^2 + y^2 - 4x - 2y + 4 = 0$

Sol. (a) Let O be the centre of axis of circle & bisect the intercept AB(6 units) i.e. XA = 3 units.

In ΔXOA, $OX^2 + XA^2 = OA^2$ [Pythagoras theorem]

$\Rightarrow 9 + 9 = OA^2$

$OA = \text{radius} = \sqrt{18}$ ...(1)

Centre of circle $+\left(3, 3\sqrt{2}\right)$

Equation : $\left(x-3\right)^2 + \left(y - 3\sqrt{2}\right)^2 = 18$

(b) Similarly in ΔXOB

$OX^2 + XB^2 = OB^2$ [Pythagoras]

$9 + 16 = OB^2$

$OB = r = 5$ ...(2)

Center of cirlce = (–5,–3)

Equation : $(x + 5)^2 + (y + 3)^2 = 25$

(c) Let O be the centre of circle with coordinates (h, k). The circle then touches x-axis at (h, 0).

Thus, radius r = k.

As, (h,k) be $x + y = 3$

$\Rightarrow K = 3 - h$

Now, $AO^2 = r^2$

$(h - 1)^2 + (3 - h - 1)^2 = (3 - h)^2$

$\Rightarrow h^2 + 1 - 2k + 4 + h^2 - 4k = 9 + h^2 - 6h$

$h^2 = 4 \Rightarrow h = \pm 2$

$K = 3 - 4 = 5,1$

Thus centre can be (–2,5) or (2,1)

Required equation :

$(x + 2)^2 + (y - 5)^2 = 25$ [r = k = 5]

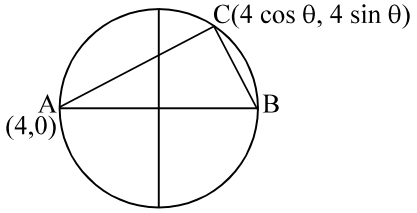
$(x - 2)^2 + (y - 1)^2 = 1$ [r = k = 1]

17. Let A (– 4, 0) and B (4, 0). Number of points C = (x, y) on the circle $x^2 + y^2 = 16$ such that the area of the triangle whose vertices are A, B and C is a positive integer, is.

माना A (– 4, 0) तथा B (4, 0) है। वृत $x^2 + y^2 = 16$ पर स्थित बिन्दुओं C = (x, y) की संख्या की त्रिभुज A, B तथा C का क्षेत्रफल पूर्णांक है।

Ans. 62

Sol. $A = \frac{1}{2} \cdot 8 \cdot 4 \sin\theta = |16 \sin\theta|$



now $\sin\theta$ can be $= \frac{1}{16}, \frac{2}{16}, \ldots\ldots \frac{15}{16}$

i.e. 15 points in each quadrant $\Rightarrow 60 + 2$ more with $\sin\theta = 1$

$\Rightarrow$ total = 62

18. If the points $(\lambda, -\lambda)$ lies inside the circle $x^2 + y^2 - 4x + 2y - 8 = 0$, then find the range of λ.

यदि बिन्दु $(\lambda, -\lambda)$ वृत $x^2 + y^2 - 4x + 2y - 8 = 0$ के अन्दर स्थित है, तो λ का परिसर ज्ञात करें।

Ans. $\lambda \in (-1, 4)$

Sol. Since the point (a, –a) lies inside the circle $x^2 + y^2 - 4x + 2y - 8 = 0$

$\therefore a^2 + a^2 - 4a - 2a - 8 < 0$

$\Rightarrow 2a^2 - 6a - 8 < 0$

$\Rightarrow a^2 - 3a - 4 < 0$

$\Rightarrow (a - 4)(a + 1) < 0 \Rightarrow -1 < a < 4$

$\Rightarrow a \in (-1,4)$

19. The circle $x^2 + y^2 - 6x - 10y + c = 0$ does not touch or intersect the coordinate axes and the point (1, 4) is inside the circle. Find the set of the values of c.

वृत $x^2 + y^2 - 6x - 10y + c = 0$ यदि अक्षों को ना तो काटता है एवं ना ही स्पर्श करता है तथा बिन्दु (1, 4) उसके अन्दर स्थित है। c का समुच्चय ज्ञात कीजिए।

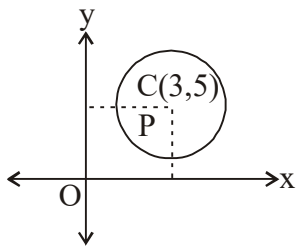
Ans. $c \in (25, 29)$

Sol. $x^2 + y^2 - 6x - 10y + c = 0$

Centre is (3,5) and radius

$= \sqrt{3^2 + 5^2 - c} = \sqrt{34 - c}$

$34 - c > 0 \Rightarrow c < 34$ ...(1)



Since, P(1,4) lies inside the circle,

$CP < r$

$\Rightarrow (3 - 1)^2 + (5 - 4)^2 < 34 - c$

$\Rightarrow c < 29$ ...(2)

Also, since the circle does not ouch or intersect the coordinate axes, we have

$r < 5$ and $r < 3$

$\Rightarrow r < 3$

$\Rightarrow \sqrt{34 - c} < 3$

$\Rightarrow c > 25$ ...(3)

From (1), (2) and (3),

$25 < c < 29$

Possible integral value of c is 26,27,28

Special DPP on Circle **DPP # 3**

1. A straight line with slope 2 and y-intercept 10 touches the circle, $x^2 + y^2 + 16x + 12y + c = 0$ at a point Q for :

(A) no value of c (B) unique value of c

(C) two values of c (D) infinite values of c

एक सरल रेखा का ढाल 2 है तथा y-अन्तःखण्ड 10 है जो वृत $x^2 + y^2 + 16x + 12y + c = 0$ को किसी बिन्दु पर स्पर्श करती है :

(A) c का कोई वास्तविक मान नही (B) c के अद्वितीय मान के लिये

(C) c के दो मानों के लिये (D) c के अनन्त मानों के लिये

Ans. A

Sol. Centre $-8, -6$; line : $y = 2x + 10$; Line passes through the centre hence for any value of c it can not touch the circle. Also perpendicular distance from centre of the circle on the line is zero $\Rightarrow$ line is a diameter

2. The smallest distance between the circle $(x - 5)^2 + (y + 3)^2 = 1$ and the line $5x + 12y - 4 = 0$, is :

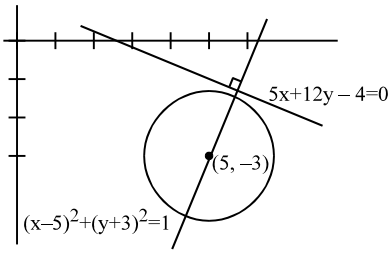
वृत $(x - 5)^2 + (y + 3)^2 = 1$ तथा रेखा $5x + 12y - 4 = 0$ के मध्य न्यूनतम दूरी है :

(A) $\frac{1}{13}$ (B) $\frac{2}{13}$ (C) $\frac{3}{15}$ (D) $\frac{4}{15}$

Ans. B

Sol. The distance from a point (x_0, y_0) to a line $Ax + By + C = 0$ is

$$\frac{|Ax_0 + By_0 + C|}{\sqrt{A^2 + B^2}}$$



The centre of the circle is $(5, -3)$,
so the distance from this point to the line $5x + 12y - 4 = 0$ is

$$\frac{5 \cdot 5 + 12 \cdot (-3) - 4}{\sqrt{5^2 + 12^2}} = \frac{|25 - 36 - 4|}{\sqrt{169}} = \frac{15}{13}$$

From this, subtract the radius of the circle, one, to obtain the answer of $\frac{2}{13}$

3. The value of 'c' for which the set, $\{(x, y) \mid x^2 + y^2 + 2x \le 1\} \cap \{(x, y) \mid x - y + c \ge 0\}$ contains only one point in common is :

'c' के मानों का समुच्चय जिसके लिये समुच्चय $\{(x, y) \mid x^2 + y^2 + 2x \le 1\} \cap \{(x, y) \mid x - y + c \ge 0\}$ में केवल एक उभयनिष्ठ बिन्दु है :

(A) $(-\infty, -1] \cup [3, \infty)$ (B) $\{-1, 3\}$ (C) $\{-3\}$ (D) $\{-1\}$

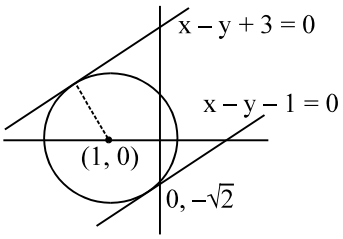
Ans. D

Sol. $x^2 + y^2 + 2x - 1 = 0$; centre $(-1, 0)$ and rad. $= \sqrt{2}$

line $x - y + c = 0$

$\left|\frac{-1+c}{\sqrt{2}}\right| = \sqrt{2}$; $|c - 1| = 2$; $c - 1 = \pm 2$

$\Rightarrow$ $c = 3$ or -1



4. The shortest distance from the line $3x + 4y = 25$ to the circle $x^2 + y^2 = 6x - 8y$ is equal to :

रेखा $3x + 4y = 25$ की वृत $x^2 + y^2 = 6x - 8y$ से न्यूनतम दूरी है :

(A) 7/5 (B) 9/5 (C) 11/5 (D) 32/5

Ans. A

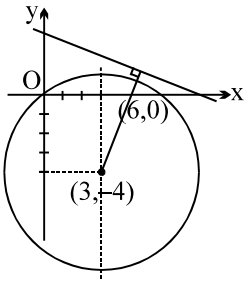
Sol. Centre: $(3, -4)$ and $r = 5$

perpendicular distance from $(3, -4)$ on

$3x + 4y - 25 = 0$ is

$p = \left|\frac{9 - 16 - 25}{5}\right| = \frac{32}{5}$

$d = \frac{32}{5} - 5 = \frac{7}{5}$ Ans.



5. If the curve $y = 1 + \sqrt{4 - x^2}$ and the line $y = (x - 2)k + 4$ has two distinct points of intersection then the range of k, is :

यदि वक्र $y = 1 + \sqrt{4 - x^2}$ तथा रेखा $y = (x - 2)k + 4$ दो अलग–अलग बिन्दुओं पर प्रतिच्छेद करते है तो k का परिसर होगा:

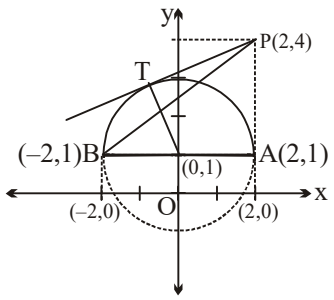
(A) $[1, 3]$ (B) $\left[\frac{5}{12}, \infty\right)$ (C) $\left[\frac{5}{12}, \frac{3}{4}\right]$ (D) $\left(\frac{5}{12}, \frac{3}{4}\right]$

Ans. D

Sol. Given $y = 1 + \sqrt{4 - x^2}$

Now curve is $(y-1)^2 = 4 - x^2 \Rightarrow x^2 + (y-1)^2 = 4$

Also line is $y - 4 = k(x-2)$, passing through (2, 4) with slope k.



Now $$m_{PB} = \frac{4-1}{2+2} = \frac{3}{4}$$

Again for slope of PT, using condition of tangency we get

$$\left|\frac{-1+4-2k}{\sqrt{1+k^2}}\right| = 2 \Rightarrow (3-2k)^2 = 4(1+k^2) \Rightarrow 9 - 12k = 4 \Rightarrow k = \frac{5}{12}$$

$\therefore$ For two distinct points of intersection, we must have $\frac{5}{12} < k \le \frac{3}{4}$. Ans.

6. Consider 3 non collinear points A, B, C with coordinates (0, 6), (5, 5) and (–1, 1) respectively. Equation of a line tangent to the circle circumscribing the triangle ABC and passing through the origin is :

माना 3 बिन्दु A, B, C क्रमशः (0, 6), (5, 5) तथा (–1, 1) है मूल बिन्दु से त्रिभुज ABC के परिवृत पर स्पर्श रेखा का समीकरण ज्ञात करें :

(A) $2x - 3y = 0$ (B) $3x + 2y = 0$ (C) $3x - 2y = 0$ (D) $2x + 3y = 0$

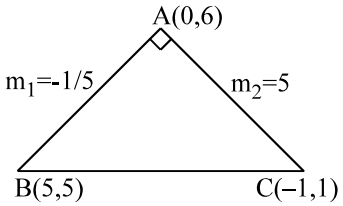
Ans. D

Sol. Note that the ABC is right angled at A

$\therefore$ equation of circle

$(x+1)(x-5) + (y-1)(y-5) = 0$

$x^2 + y^2 - 4x - 6y - 5 + 5 = 0$



$x^2 + y^2 - 4x - 6y = 0$

Hence the circle passes through the origin

$\therefore$ tangent at (0, 0) is $4x + 6y = 0$

$2x + 3y = 0$

7. A straight line with slope 2 and y-intercept 5 touches the circle, $x^2 + y^2 + 16x + 12y + c = 0$ at a point Q. Then the coordinates of Q are :

एक सरल रेखा जिसकी ढाल 2 तथा y-अन्तःखण्ड 5 है वृत $x^2 + y^2 + 16x + 12y + c = 0$ को किसी बिन्दु Q पर स्पर्श करती है तो

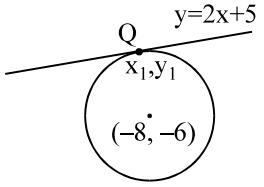
Q के निर्देशांक है :

(A) (–6, 11) (B) (–9, –13) (C) (–10, – 15) (D) (–6, –7)

Ans. D

Sol. $y_1 = 2x_1 + 5$ and $\frac{(y_1+6)}{x_1+8} \times 2 = -1$

$\Rightarrow x_1 = -6$ and $y_1 = -7$



8. The line $2x - y + 1 = 0$ is tangent to the circle at the point (2, 5) and the centre of the circles lies on $x - 2y = 4$. The radius of the circle is :

रेखा $2x - y + 1 = 0$ एक वृत पर बिन्दु (2, 5) पर स्पर्श रेखा है तथा केन्द्र $x - 2y = 4$ पर है। वृत की त्रिज्या है :

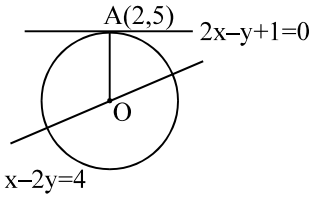
(A) $3\sqrt{5}$ (B) $5\sqrt{3}$ (C) $2\sqrt{5}$ (D) $5\sqrt{2}$

Ans. A

Sol. $2x - y + 1 = 0$ is tangent

slope of line OA $= -\frac{1}{2}$

equation of OA, $(y-5) = -\frac{1}{2}(x-2)$



$2y - 10 = -x + 2$

$x + 2y = 12$

$\therefore$ intersection with $x - 2y = 4$ will give coordinates of centre

solving we get (8, 2)

distance OA $= \sqrt{(8-2)^2 + (2-5)^2} = \sqrt{36+9} = \sqrt{45} = 3\sqrt{5}$

9. To which of the following circles, the line $y - x + 3 = 0$ is normal at the point $\left(3 + \frac{3}{\sqrt{2}}, \frac{3}{\sqrt{2}}\right)$:

निम्न में से किन वृतो पर रेखा $y - x + 3 = 0$ बिन्दु $\left(3 + \frac{3}{\sqrt{2}}, \frac{3}{\sqrt{2}}\right)$ पर अभिलम्ब है :

(A) $\left(x - 3 - \frac{3}{\sqrt{2}}\right)^2 + \left(y - \frac{3}{\sqrt{2}}\right)^2 = 9$ (B) $\left(x - \frac{3}{\sqrt{2}}\right)^2 + \left(y - \frac{3}{\sqrt{2}}\right)^2 = 9$

(C) $x^2 + (y-3)^2 = 9$ (D) $(x-3)^2 + y^2 = 9$

Ans. D

Sol. Line must pass through the centre of the circle

10. The equation of a line which is concurrent with lines $L_1 : x - 2y - 1 = 0$ and $L_2 : x + y - 2 = 0$ and tangent to the circle $x^2 + y^2 - 6x + 2y + 6 = 0$, is :

(A) $3x - 3y = 4$ (B) $9x - 3y = 14$ (C) $3x - 9y = 2$ (D) none

रेखाओं $L_1 : x - 2y - 1 = 0$ तथा $L_2 : x + y - 2 = 0$ से संगामी तथा वृत $x^2 + y^2 - 6x + 2y + 6 = 0$ पर स्पर्शी रेखा का समीकरण है :

(A) $3x - 3y = 4$ (B) $9x - 3y = 14$ (C) $3x - 9y = 2$ (D) कोई नही

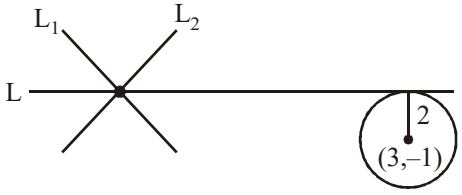
Ans. D

Sol. The line is $(x - 2y - 1) + \lambda (x + y - 2) = 0$ ....(1)

$\Rightarrow x(1 + \lambda) + y(\lambda - 2) - (1 + 2\lambda) = 0$

For the circle Centre $\equiv (3, -1)$

Radius = $\sqrt{9 + 1 - 6} = 2$



$\because$ $\perp$, distance of line from the centre of circle is equal to radius.

$\therefore \quad \dfrac{|3(1+\lambda) - (\lambda - 2) - (1 + 2\lambda)|}{\sqrt{(1+\lambda)^2 + (\lambda - 2)^2}} = 2$

$\Rightarrow (3 + 3\lambda - \lambda + 2 - 1 - 2\lambda)^2 = 4[1 + \lambda^2 + 2\lambda + \lambda^2 - 4\lambda + 4]$

$\Rightarrow 4^2 = 4(2\lambda^2 - 2\lambda + 5)$ $\Rightarrow 2\lambda^2 - 2\lambda + 1 = 0$

$\because$ $D < 0 \Rightarrow$ No real λ

$\therefore$ No such line exists

Alternate : - $L_1 : x - 2y = 1$ .....(1)

$L_2 : x + y = 2$ .....(2)

Point of intersection : $\left(\dfrac{5}{3}, \dfrac{1}{3}\right)$ which lies inside the circle

$\because \quad S_1 = \left(\dfrac{5}{3}\right)^2 + \left(\dfrac{1}{3}\right)^2 - 6 \times \left(\dfrac{5}{3}\right) + 2 \times \left(\dfrac{1}{3}\right) + 6 = \dfrac{-4}{9} < 0$

There can't be a tangent through an interior point of circle.

11. The radius of the circle whose two normals are represented by the equation $x^2 - 5xy - 5x + 25y = 0$ and which touches externally the circle $x^2 + y^2 - 2x + 4y - 4 = 0$ will be :

उस वृत की त्रिज्या जिसके दो अभिलम्ब समीकरणों $x^2 - 5xy - 5x + 25y = 0$ द्वारा व्यक्त किये जाते है तथा जो वृत $x^2 + y^2 - 2x + 4y - 4 = 0$ को बाह्य स्पर्श करता है :

(A) 21 (B) 2 (C) 3 (D) 14

Ans. B

Sol. Normals are $x^2 - 5xy - 5x + 25y = 0$ which can be factorised as, $(x - 5)(x - 5y) = 0$

Hence $x = 5$ and $y = 1 \Rightarrow$ centre of the circle is $(5, 1)$

and circle $x^2 + y^2 - 2x + 4y - 4 = 0$ is

$(x - 1)^2 + (y + 2)^2 = 3^2$

centre $(1, -2)$ and radius 3

external touch $C_1C_2 = r_1 + r_2$

$5 = 3 + \lambda$

so $\lambda = 2$ Ans. $\Rightarrow$ B

12. The normal at the point (3, 4) on a circle cuts the circle at the point (–1, –2). Then the equation of the circle is :

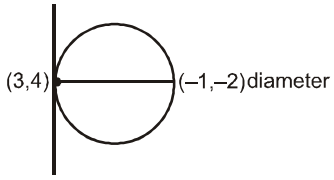
वृत पर बिन्दु (3, 4) से एक अभिलम्ब वृत को (–1, –2) पर काटता है। तो वृत का समीकरण है :

(A) $x^2 + y^2 + 2x - 2y - 13 = 0$ (B) $x^2 + y^2 - 2x - 2y - 11 = 0$

(C) $x^2 + y^2 - 2x + 2y + 12 = 0$ (D) $x^2 + y^2 - 2x - 2y + 14 = 0$

Ans. B

Sol.



$(x - 3)(x + 1) + (y - 4)(y + 2) = 0$

Equation $x^2 + y^2 - 2x - 2y - 11 = 0$

13. The tangent(s) from the point of intersection of the lines $2x - 3y + 1 = 0$ and $3x - 2y - 1 = 0$ to the circle $x^2 + y^2 + 2x - 4y = 0$ is(are) :

रेखाओं $2x - 3y + 1 = 0$ तथा $3x - 2y - 1 = 0$ के प्रतिच्छेदन बिन्दु से वृत $x^2 + y^2 + 2x - 4y = 0$ पर स्पर्श रेखाओं का समीकरण है :

(A) $x + 2y = 0, x - 2y + 1 = 0$ (B) $2x - y - 1 = 0$

(C) $y = x, y = 3x - 2$ (D) $2x + y + 1 = 0$

Ans. B

Sol. $2x - 3y + 1 = 0$ ....(1)

$3x - 2y - 1 = 0$ ....(2)

circle $x^2 + y^2 + 2x - 4y = 0$

intersection of (1) and (2) is (1, 1) (which lies on the circle)

Tangent from (1, 1) to circle

$x + y + (x + 1) - 2(y + 1) = 0$

$2x - y - 1 = 0$

14. A foot of the normal from the point (4, 3) to a circle is (2, 1) and a diameter of the circle has the equation $2x - y - 2 = 0$. Then the equation of the circle is :

बिन्दु (4, 3) से वृत पर अभिलम्ब का पाद (2, 1) है तथा वृत का व्यास $2x - y - 2 = 0$ है तो वृत का समीकरण है :

(A) $x^2 + y^2 - 4y + 2 = 0$ (B) $x^2 + y^2 - 4y + 1 = 0$ (C) $x^2 + y^2 - 2x - 1 = 0$ (D) $x^2 + y^2 - 2x + 1 = 0$

Ans. C

Sol. Diameter of the circle is given as $2x - y - 2 = 0$ ....(1)

slope of PN: $\frac{3-1}{4-2} = 1$

equation of normal through PN is

$y - 1 = (x - 2)$

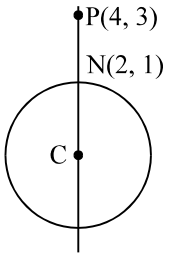
$x - y - 1 = 0$ ....(2)

solving (1) and (2) centre is (1, 0)

hence equation of the circle is

$(x - 1)^2 + y^2 = (2 - 1)^2 + 1$

$x^2 + y^2 - 2x - 1 = 0$



15. The circle having $2x + y - 5 = 0$ and $2x + y + 15 = 0$ as tangents and (–5, –5) is one of the point of contact of one of them, then the equation of circle is :

वृत जिसकी स्पर्श रेखाऐं $2x + y - 5 = 0$ तथा $2x + y + 15 = 0$ एवं उनमें से एक का स्पर्शी बिन्दु (–5, –5) है तो वृत का समीकरण ज्ञात करें :

(A) $x^2 + y^2 + 6x + 4y - 15 = 0$ (B) $x^2 + y^2 + 4x + 2y - 95 = 0$

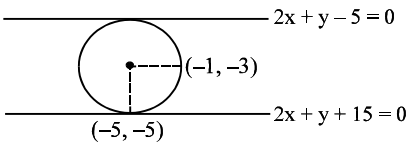
(C) $x^2 + y^2 + 2x + 6y - 10 = 0$ (D) $x^2 + y^2 + 6x + 4y - 105 = 0$

Ans. C

Sol. Diameter of circle is distance between the 2 parallel lines $= \frac{20}{\sqrt{5}} = 4\sqrt{5}$

slope of the normal at $(-5,-5)$ is $\tan\theta = \frac{1}{2}$ (–ve reciprocal of tangent)

equation of normal $\frac{x+5}{\cos\theta} = \frac{y+5}{\sin\theta} = 2\sqrt{5}$



$x = 2\sqrt{5}\cos\theta - 5$, $y = 2\sqrt{5}\sin\theta - 5$

$\Rightarrow$ $x = -1$, $y = -3$ (this is the centre)

equation of circle $(x+1)^2 + (y+3)^2 = 20$

16. Sum of all the radii of the circles touching the coordinate axes and the line $3x + 4y = 12$, is :

उन सभी वृत्तों की त्रिज्याओं का योग क्या होगा जो रेखा $3x + 4y = 12$ एवं निर्देशांक अक्षों को स्पर्श करती है :

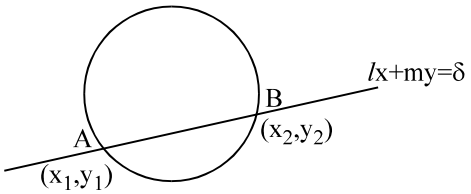
(A) 1 (B) 2 (C) 12 (D) $\frac{9}{2}$

Ans. C

Sol. Solving line and circle

$m^2x^2 + (\delta - lx)^2 = m^2c^2$

$(m^2 + l^2)x^2 - 2l\delta x + (\delta^2 - m^2c^2) = 0 \lessdot^{x_1}_{x_2}$ $(l^2 + m^2 = 1)$



given $H = \frac{2x_1x_2}{x_1 + x_2} = \frac{2(\delta^2 - m^2c^2)}{2l\delta}$

$\Rightarrow$ $lH = \frac{\delta^2 - m^2c^2}{\delta}$ ....(1)

|||ly $mK = \frac{\delta^2 - c^2l^2}{\delta}$ ....(2)

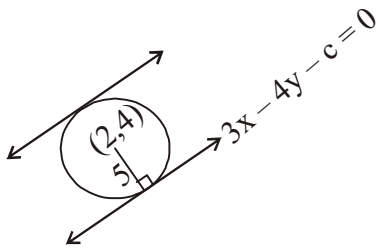
(1) + (2) $lH + mK = \frac{2\delta^2 - c^2(l^2 + m^2)}{\delta} = 2\delta - \frac{c^2}{\delta}$ where $(l^2 + m^2 = 1)$

17. Show that the line $3x - 4y - c = 0$ will meet the circle having centre at (2, 4) and the raidus 5 in real and distinct points if $-35 < c < 15$.

दर्शाइये कि रेखा $3x - 4y - c = 0$ वृत जिसका केन्द्र (2, 4) एवं त्रिज्या 5 है को भिन्न वास्तविक बिन्दुओं पर मिल सकती है यदि $-35 < c < 15$ है।

Ans.

Sol.



For the line to meet the circle in two distinct points,

$$\left|\frac{6-16-c}{\sqrt{25}}\right| < 5$$

$-5 < -c - 10 < 25$

$-15 < -c < 35$

$-35 < c < 15$

18. Find the equation of the tangent to the circle :

(a) $x^2 + y^2 - 6x + 4y = 12$, which are parallel to the straight line $4x + 3y + 5 = 0$.

(b) $x^2 + y^2 - 22x - 4y + 25 = 0$, which are perpendicular to the straight line $5x + 12y + 9 = 0$.

(c) $x^2 + y^2 = 25$, which are inclined at 30° to the axis of x.

वृत पर स्पर्श रेखा का समीकरण ज्ञात करें :

(a) $x^2 + y^2 - 6x + 4y = 12$ पर की रेखा $4x + 3y + 5 = 0$ के समानान्तर है

(b) $x^2 + y^2 - 22x - 4y + 25 = 0$ की रेखा $5x + 12y + 9 = 0$ पर लम्बवत् है

(c) $x^2 + y^2 = 25$ की x अक्ष से 30° नत है

Ans. (a) $4x + 3y + 19 = 0$ and $4x + 3y - 31 = 0$ (b) $12x - 5y + 8 = 0$ and $12x - 5y - 252 = 0$

(c) $x - \sqrt{3}\,y \pm 10 = 0$

Sol. (a) Let equation of tangent be $4x + 3y + k = 0$, then $\sqrt{9+4+12} = \left|\frac{4(3)+3(-2)+k}{\sqrt{16+9}}\right|$

$\Rightarrow 6 + k = \pm 25 \Rightarrow k = 19$ and -31.

Hence the tangents are $4x + 3y + 19 = 0$ and $4x + 3y - 31 = 0$.

(b) Given circle is

$x^2 + y^2 - 22x - 4y + 25 = 0$

Centre is (11,2) and radius is

$\sqrt{121 + 4 - 25} = 10$

Equation of line perpendicular to $5x + 12y + 8 = 0$ is $12x - 5y + k = 0$.

If this line touches the circle, then distance of the centre of the circle from the line radius.

$$\therefore \left|\frac{12(11)-5(2)+k}{\sqrt{144+25}}\right| = 10$$

$\Rightarrow |k + 122| = 130$

$\Rightarrow k = 8$ or -252

(c) $y = mx \pm a\sqrt{1+m^2}$

$$y = \frac{x}{\sqrt{3}} \pm 5\sqrt{1+\frac{1}{3}}$$

$\Rightarrow x - \sqrt{3}y \pm 10 = 0$

19. Find the equations of the tangents to the circle $x^2 + y^2 - 6x - 4y + 5 = 0$, which make an angle of 45° with the axis of x.

वृत $x^2 + y^2 - 6x - 4y + 5 = 0$ पर उन स्पर्श रेखाओं का समीकरण ज्ञात कीजिये जो x अक्ष के साथ 45° कोण बनाती है।

Ans. $y = x + 3$ and $y = x - 5$

Sol. Since the slope of the tangent to the circle is 45° , its equation can be taken as $y = x + c$.
The equation of the circle is

$x^2 + y^2 - 6x - 4y + 5 = 0$, i.e., $(x - 3)^2 + (y - 2)^2 = \left(\sqrt{8}\right)^2$.

The centre of the circle is $(3, 2)$ and its radius $\sqrt{8}$.

The length of the perpendicular to $y = x + c$ from $(3, 2)$ = radius.

$$\therefore \quad \pm \frac{2-3-c}{\sqrt{2}} = \sqrt{8}.$$

i.e., $c = 3, -5$.

$\therefore$ The tangents are $y = x + 3$ and $y = x - 5$

20. The lines $lx + my + n = 0$ intersects the curve $ax^2 + by^2 = 1$ $(a \neq b)$ in A and B. If AOB = 90°, that AB touches the circle say C. Find the equation of C.

रेखायें $lx + my + n = 0$ वक्र $ax^2 + by^2 = 1$ $(a \neq b)$ को बिन्दुओं A तथा B पर प्रतिच्छेद करती है। यदि कोण AOB = 90° है, AB वृत C को स्पर्श करती है। वृत C का समीकरण ज्ञात करें।

Ans. $(a + b)(x^2 + y^2) = 1$

Sol. $lx + my + n = 0 \Rightarrow \text{H.F.} = \left(\frac{lx + my}{-n}\right)$

$ax^2 + by^2 = 1 \qquad (a \neq b)$

Combined equation of OA & OB :-

$$ax^2 + by^2 = \frac{(lx + my)^2}{n^2}$$

$an^2x^2 + bn^2y^2 = l2x^2 + m^2y^2 + 2lmxy$

$(l^2 - an^2)x^2 + (m^2 - bn^2)y^2 + 2lmxy = 0$

$OA \perp OB \Rightarrow l^2 - an^2 + m^2 - bn^2 = 0$

$l^2 + m^2 = (a + b)\, n^2$

$$\Rightarrow \frac{n^2}{\ell^2 + m^2} = \frac{1}{a + b}$$

$$\Rightarrow \left|\frac{n}{\sqrt{\ell^2 + m^2}}\right| = \frac{1}{\sqrt{a + b}}$$

$p = r$

centre of circle is (0,0)

and radius $= \frac{1}{\sqrt{a + b}}$

$$x^2 + y^2 = \frac{1}{a + b}$$

21. A line with gradient 2 is passing through the point P(1, 7) and touches the circle $x^2 + y^2 + 16x + 12y + c = 0$ at the point Q. If (a, b) are the coordinates of the point Q, then find the value of $(7a + 7b + c)$.

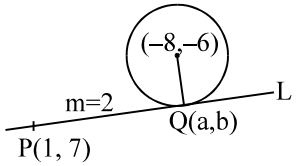
एक रेखा जिसकी प्रवणता दो है तथा बिन्दु P(1, 7) से गुजरती है तथा वृत को $x^2 + y^2 + 16x + 12y + c = 0$ पर Q को स्पर्श करती है। यदि (a, b) बिन्दु Q के निर्देशांक है तो $(7a + 7b + c) = ?$

Ans. 4

Sol. $L : y - 7 = 2(x - 1) \Rightarrow y = 2x + 5$

Now $\left(\frac{b+6}{a+8}\right)(2) = -1 \Rightarrow 2b + 12 = -a - 8 \Rightarrow a + 2b = -20$ .....(1)

Also (a, b) lies on L, so we get



$b = 2a + 5$ .....(2)

$\therefore$ From (1) and (2), we get

$a = -6,\ b = -7$

Hence Q is (–6, – 7)

Also perpendicular from (–8, – 6) on L = r, gives [12th, 20-06-2010 P-2]

$$\left|\frac{(-16)+6+5}{\sqrt{5}}\right| = \sqrt{64+36-c} \Rightarrow \left|-\sqrt{5}\right| = \sqrt{100-c} \Rightarrow 5 = 100 - c \Rightarrow c = 95$$

Hence $(7a + 7b + c) = -42 - 49 + 95 = 4$

22. If M and m are the maximum and minimum values of $\frac{y}{x}$ for pair of real number (x, y) which satisfy the equation $(x-3)^2 + (y-3)^2 = 6$, then find the value of (M + m).

यदि M तथा m वास्तविक संख्या युग्म (x, y) के लिये $\frac{y}{x}$ के क्रमशः अधिकतम एवं न्यूनतम मान है, जो समीकरण $(x-3)^2 + (y-3)^2 = 6$ को संतुष्ट करते है, तो (M + m) का मान ज्ञात कीजिए।

Ans. 6

Sol. Let $\frac{y}{x} = m'$. So on putting $y = m'x$ in the given circle, we get

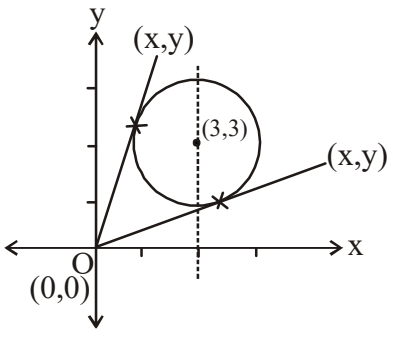
$(x-3)^2 + (m'x-3)^2 = 6 \Rightarrow x^2(1+m'^2) - 6x(1+m') + 12 = 0$

Putting discriminant = 0

$\therefore\ m' = 3 \pm 2\sqrt{2}$

So $m = 3 - 2\sqrt{2}$ and $M = 3 + 2\sqrt{2}$

Hence (M + m) = 6



1. Let ABC be a triangle with $\angle A = 45°$. Let P be a point on the side BC with PB = 3 and PC = 5. If 'O' is the circumcentre of the triangle ABC then the length OP is equal to :

माना ABC एक त्रिभुज है, $\angle A = 45°$. मान P भुजा BC पर एक बिन्दु है जहाँ PB = 3 एवं PC = 5 है। यदि 'O' त्रिभुज ABC का परिकेन्द्र है तो OP की लम्बाई है :

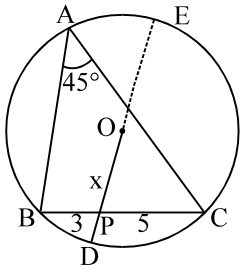
(A) $\sqrt{15}$ (B) $\sqrt{17}$ (C) $\sqrt{18}$ (D) $\sqrt{19}$

Ans. B

Sol. Using sine law

$$\frac{a}{\sin A} = 2R$$

$$8\sqrt{2} = 2R \quad \Rightarrow \quad R = 4\sqrt{2}$$



using power of a point

$$(PB)(PC) = (PD)(PE)$$

$$15 = (R - x)(R + x)$$

$$15 = R^2 - x^2 \quad \Rightarrow \quad x^2 = R^2 - 15 = 32 - 15 = 17$$

$\therefore$ $x = \sqrt{17}$ Ans.

2. In the xy-plane, the length of the shortest path from (0, 0) to (12, 16) that does not go inside the circle $(x - 6)^2 + (y - 8)^2 = 25$ is :

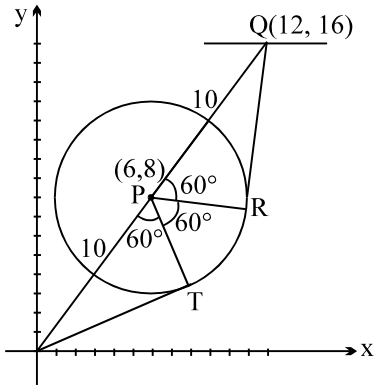
xy तल में, (0, 0) से (12, 16) के मध्य न्यूनतम दूरी का पथ की लम्बाई जो वृत $(x - 6)^2 + (y - 8)^2 = 25$ के अन्दर से नहीं जाता है, होगी :

(A) $10\sqrt{3}$ (B) $10\sqrt{5}$ (C) $10\sqrt{3} + \frac{5\pi}{3}$ (D) $10 + 5\pi$

Ans. C

Sol. Let O = (0, 0), P = (6, 8) and Q = (12, 16).

As shown in the figure the shortest route consists of tangent OT, minor arc TR and tangent RQ.
Since OP = 10, PT = 5, and $\angle OTP = 90°$,
it follows that $\angle OPT = 60°$ and $OT = 5\sqrt{3}$.
By similar reasoning, $\angle QPR = 60°$ and $QR = 5\sqrt{3}$.
Because O, P and Q are collinear (why?),
$\angle RPT = 60°$, so arc TR is of length $\frac{5\pi}{3}$.
Hence the length of the shortest route is $2(5\sqrt{3}) + \frac{5\pi}{3}$ Ans.



3. A variable circle C has the equation $x^2 + y^2 - 2(t^2 - 3t + 1)x - 2(t^2 + 2t)y + t = 0$, where t is a parameter. If the power of point P(a,b) w.r.t. the circle C is constant then the ordered pair (a, b) is :

एक चर वृत जिसका समीकरण $C : x^2 + y^2 - 2(t^2 - 3t + 1)x - 2(t^2 + 2t)y + t = 0$, है, जहाँ t प्राचल है। यदि बिन्दु P(a,b) की वृत C के सापेक्ष बिन्दु क्षमता नियत है तो क्रमित युग्म (a, b) होना चाहिये :

(A) $\left(\frac{1}{10}, -\frac{1}{10}\right)$ (B) $\left(-\frac{1}{10}, \frac{1}{10}\right)$ (C) $\left(\frac{1}{10}, \frac{1}{10}\right)$ (D) $\left(-\frac{1}{10}, -\frac{1}{10}\right)$

Ans. B

Sol. Power : $P = a^2 + b^2 - 2(t^2 - 3t + 1)a - 2(t^2 + 2t)b + t$
$= -(2a + 2b)t^2 + (6a - 4b + 1)t + a^2 + b^2 - 2a$
This power is independent of the parameter t
if and only if $2a + 2b = 0 \Rightarrow a = -b$
and $6a - 4b + 1 = 0$
$\Rightarrow a = -\frac{1}{10}$ and $b = \frac{1}{10}$

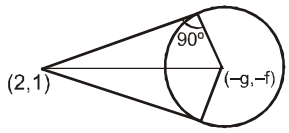
4. A point (2, 1) is outside the circle $x^2+y^2+2gx+2fy+c=0$ and AP, AQ are tangents to the circle. The equation of the circle circumscribing the triangle APQ is :

एक बिन्दु A(2, 1) वृत $x^2 + y^2 + 2gx + 2fy + c = 0$ के बाहर है तथा AP, AQ दो स्पर्श रेखायें है। त्रिभुज APQ का परिवृत का समीकरण है :

(A) $(x+g)(x-2)+(y+f)(y-1)=0$ (B) $(x+g)(x-2)-(y+f)(y-1)=0$
(C) $(x-g)(x+2)+(y-f)(y+1)=0$ (D) none

Ans. A

Sol.



$(x + g)(x - 2) + (y + f)(y - 1) = 0$

5. Tangents are drawn from (4, 4) to the circle $x^2 + y^2 - 2x - 2y - 7 = 0$ to meet the circle at A and B. The length of the chord AB is :

बिन्दु (4, 4) से वृत $x^2 + y^2 - 2x - 2y - 7 = 0$ पर डाली गयी स्पर्श रेखायें वृत को A तथा B पर मिलती है। जीवा AB की लम्बाई है :

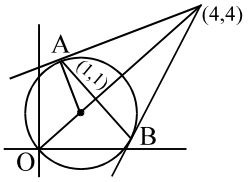
(A) $2\sqrt{3}$ (B) $3\sqrt{2}$ (C) $2\sqrt{6}$ (D) $6\sqrt{2}$

Ans. B

Sol. length of C.O.C. $= \dfrac{2RL}{\sqrt{R^2 + L^2}}$

equation of circle $x^2 + y^2 - 2x - 2y - 7 = 0$

equation of COC $= 4x + 4y - (x + 4) - (y + 4) - 7 = 0 = 3x + 3y - 15 = 0$



radius = 3 perpendicular from (1, 1) $= \left|\dfrac{6-15}{\sqrt{18}}\right| = \dfrac{9}{\sqrt{18}} = \dfrac{9}{3\sqrt{2}} = \dfrac{3}{\sqrt{2}}$

$\therefore$ length $\dfrac{AB}{2} = \sqrt{r^2 - \left(\dfrac{3}{\sqrt{2}}\right)^2} = \dfrac{3}{\sqrt{2}}$

$\therefore$ $AB = 3\sqrt{2}$

6. The area of the quadrilateral formed by the tangents from the point (4 , 5) to the circle $x^2 + y^2 - 4x - 2y - 11 = 0$ with the pair of radii through the points of contact of the tangents is :

(A) 4 sq.units (B) 8 sq.units (C) 6 sq.units (D) none

बिन्दु (4 , 5) से वृत $x^2 + y^2 - 4x - 2y - 11 = 0$ पर डाली गयी स्पर्श रेखाओं तथा स्पर्शी बिन्दुओं के सापेक्ष त्रिज्या युग्म द्वारा बनाये गये चतुर्भुज का क्षेत्रफल है :

(A) 4 वर्ग इकाई (B) 8 वर्ग इकाई (C) 6 वर्ग इकाई (D) कोई नहीं

Ans. B

Sol. $C : x^2 + y^2 - 4x - 2y - 11 = 0$

centre (2,1)

$r = \sqrt{4 + 1 + 11} = 4$

Area of the Quadrilateral PACB is

A = (Radius) × (Length of tangent)

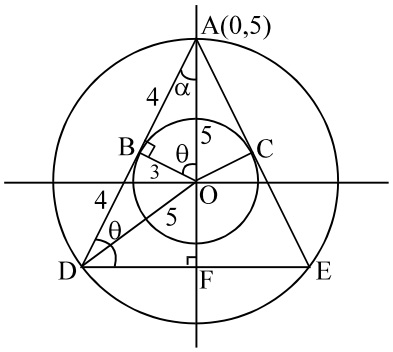
7. Consider the circles, $x^2 + y^2 = 25$ and $x^2 + y^2 = 9$. From the point A (0, 5) two segments are drawn touching the inner circle at the points B and C while intersecting the outer circle at the points D and E. If 'O' is the centre of both the circles then the length of the segment OF that is perpendicular to DE, is :

दिया हुआ वृत्त $x^2 + y^2 = 25$ तथा $x^2 + y^2 = 9$ है बिन्दु A (0, 5) से दो रेखा खण्ड खींचा गया है जो कि आन्तरिक वृत्त को B एवं C पर स्पर्श करता है तथा बाह्य वृत को D एवं E पर काटता है जहाँ वृत्त के केन्द्र 'O' है तो DE पर लम्ब रेखाखण्ड OF की लम्बाई होगी:

(A) 7/5 (B) 7/2 (C) 5/2 (D) 3

Ans. A

Sol. Triangles ABO and ADF are similiar



$$\frac{AB}{AF} = \frac{AO}{AD}$$

$$\frac{4}{AF} = \frac{5}{8} \quad \Rightarrow \quad AF = \frac{32}{5}$$

Hence OF = $\frac{32}{5} - 5 = \frac{7}{5}$

8. If a circle of radius 3 units is touching the lines $\sqrt{3}\ y^2 - 4xy + \sqrt{3}\ x^2 = 0$ in the first quadrant then the length of chord of contact to this circle, is :

त्रिज्या 3 वाली तथा रेखा युग्मो $\sqrt{3}\ y^2 - 4xy + \sqrt{3}\ x^2 = 0$ को प्रथम चतुर्थांश में स्पर्श करने वाली वृत की स्पर्श जीवा की लम्बाई होगी :

(A) $\frac{\sqrt{3}+1}{2}$ (B) $\frac{\sqrt{3}+1}{\sqrt{2}}$ (C) $3\left(\frac{\sqrt{3}+1}{\sqrt{2}}\right)$ (D) $3\frac{\left(\sqrt{3}+1\right)}{2}$

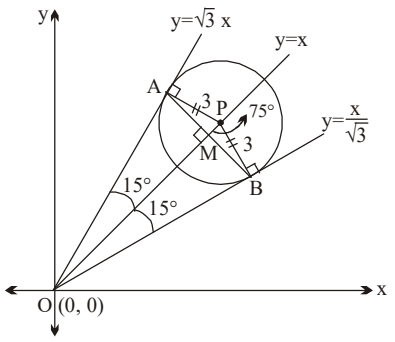
Ans. C

Sol. Given equation of lines $\sqrt{3}\ y^2 - 4xy + \sqrt{3}\ x^2 = 0$

$\Rightarrow \sqrt{3}y^2 - 3xy - xy + \sqrt{3}\,x^2 = 0$

$\Rightarrow \left(\sqrt{3}y - x\right)\left(y - \sqrt{3}\,x\right) = 0$

$\Rightarrow y = \frac{x}{\sqrt{3}}, y = \sqrt{3}\ x$



$\therefore \angle APO = 75°$

In ΔAMP, $\sin 75° = \frac{AM}{3} \Rightarrow AM = 3 \sin 75°$

Paragraph for Question nos. 9 to 11

Consider a circle $x^2 + y^2 = 4$ and a point P(4, 2). θ denotes the angle enclosed by the tangents from P on the circle and A, B are the points of contact of the tangents from P on the circle.

दिया हुआ वृत $x^2 + y^2 = 4$ है तो बिन्दु P(4, 2) है। P से वृत पर खीचीं गयी स्पर्श रेखा PA एवं PB है तथा जिनके बीच का कोण θ है।

9. The value of θ lies in the interval :

θ किस अन्तराल में होगा :

(A) (0, 15°) (B) (15°, 30°) (C) 30°, 45°) (D) (45°, 60°)

Ans. D

Sol.

10. The intercept made by a tangent on the x-axis is :

स्पर्श रेखा x-अक्ष पर कितना अन्तःखण्ड काटेगी :

(A) 9/4 (B) 10/4 (C) 11/4 (D) 12/4

Ans. B

Sol.

11. Locus of the middle points of the portion of the tangent to the circle terminated by the coordinate axes is :

यदि वृत पर बिन्दु P से खीचीं गयी स्पर्श रेखा की अक्षों के बीच के भाग के मध्य बिन्दु का बिन्दुपथ होगा :

(A) $x^{-2} + y^{-2} = 1^{-2}$ (B) $x^{-2} + y^{-2} = 2^{-2}$ (C) $x^{-2} + y^{-2} = 3^{-2}$ (D) $x^{-2} - y^{-2} = 4^{-2}$

Ans. A

Sol. Tangent

$y - 2 = m(x - 4)$

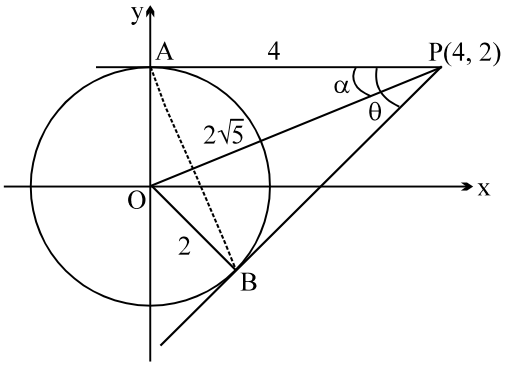
$mx - y + (2 - 4m) = 0$

$p = \left| \frac{2 - 4m}{\sqrt{1 + m^2}} \right| = 2$

$(1 - 2m)^2 = 1 + m^2$

$3m^2 - 4m = 0$

$m = 0$ or $m = \frac{4}{3}$



Hence equation of tangent is $y = 2$ and (with infinite intercept on x-axis)

or $\quad y - 2 = \frac{4}{3}(x - 4) \quad \Rightarrow \quad 3y - 6 = 4x - 16 \quad \Rightarrow \quad 4x - 3y - 10 = 0$

x-intercept $= \frac{10}{4}$ Ans.(ii) $\Rightarrow$ (B)

Variable line with mid point (h, k)

$\frac{x}{2h} + \frac{y}{2k} = 1$, it touches the circle $x^2 + y^2 = 4$

$\therefore \quad \left| \frac{-1}{\sqrt{\frac{1}{4h^2} + \frac{1}{4k^2}}} \right| = 2 \Rightarrow \frac{1}{4h^2} + \frac{1}{4k^2} = \frac{1}{4} \Rightarrow$ locus is $x^{-2} + y^{-2} = 1$ Ans.(iii) $\Rightarrow$ (A)

12. Tangents OP and OQ are drawn from the origin O to the circle $x^2 + y^2 + 2gx + 2fy + c = 0$. Find the equation of the circumcircle of the triangle OPQ.

मूल बिन्दु से वृत $x^2 + y^2 + 2gx + 2fy + c = 0$ पर दो स्पर्श रेखाऐं OP एवं OQ डाली जाती है। त्रिभुज OPQ के परिवृत की त्रिज्या ज्ञात करें।

Ans. $x^2 + y^2 + gx + fy = 0$

Sol.

13. From any point on the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ tangent are drawn to the circle $x^2 + y^2 + 2gx + 2fy + c \sin^2 \alpha + (g^2 + f^2) \cos^2 \alpha = 0$. Find the angle between them.

वृत $x^2 + y^2 + 2gx + 2fy + c = 0$ पर किसी बिन्दु से वृत $x^2 + y^2 + 2gx + 2fy + c \sin^2 \alpha + (g^2 + f^2) \cos^2 \alpha = 0$ पर डाली गयी स्पर्श रेखाओं के मध्य कोण ज्ञात कीजिए।

Ans. 2α

Sol.

14. If the length of the tangent from a point (f, g) to the circle $x^2 + y^2 = 4$ be four times the length of the tangent from it to the circle $x^2 + y^2 = 4x$, find the value of $64f - 15(g^2 + f^2)$.

यदि बिन्दु (f, g) से वृत $x^2 + y^2 = 4$ पर स्पर्श रेखा की लम्बाई वृत $x^2 + y^2 = 4x$ पर स्पर्श रेखा से चार गुना है तो $64f - 15(g^2 + f^2)$ का मान ज्ञात करें।

Ans. 4

Sol.

1. From (3, 4) chords are drawn to the circle $x^2+y^2-4x=0$. The locus of the mid points of the chords is :

(3, 4) से वृत $x^2 + y^2 - 4x = 0$ पर जीवाऐं बनाई जाती है। जीवाओं के मध्य बिन्दु का बिन्दुपथ है :

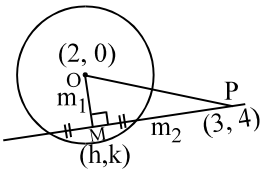
(A) $x^2 + y^2 - 5x - 4y + 6 = 0$ (B) $x^2 + y^2 + 5x - 4y + 6 = 0$

(C) $x^2 + y^2 - 5x + 4y + 6 = 0$ (D) $x^2 + y^2 - 5x - 4y - 6 = 0$

Ans. A

Sol. Arc of the circle with OP as diameter intercepted by the given circle.

$m_1m_2 = -1$



2. The locus of the center of the circles such that the point (2, 3) is the mid point of the chord $5x+2y=16$ is :

(A) $2x-5y+11=0$ (B) $2x+5y-11=0$

(C) $2x+5y+11=0$ (D) none

बिन्दु (2, 3) यदि किसी जीवा $5x+2y=16$ का मध्य बिन्दु है तो उसके सापेक्ष वृतो के केन्द्र का बिन्दुपथ ज्ञात कीजिये :

(A) $2x-5y+11=0$ (B) $2x+5y-11=0$

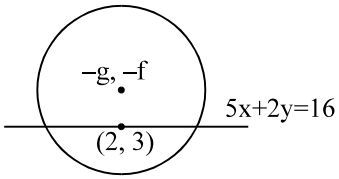
(C) $2x+5y+11=0$ (D) कोई नहीं

Ans. A

Sol. Slope of the given line $= -5/2$

$\Rightarrow -\frac{5}{2}.\frac{3+f}{2+g} = -1 \Rightarrow 15 + 5f = 4 + 2g$

$\Rightarrow$ locus is $2x - 5y + 11 = 0$



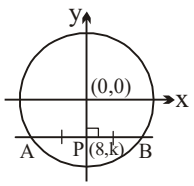
3. Number of integral values of 'k' for which the chord of the circle $x^2 + y^2 = 125$ passing through P(8, k) gets bisected at P (8, k) and has integral slope is :

'k' के पूर्णांक मानों की संख्या जिसके लिये वृत $x^2 + y^2 = 125$ के लिये बिन्दु P (8, k) से जाने वाली जीवा बिन्दु P (8, k) पर ही समद्विभाजित होती है तथा पूर्णांक प्रवणता रखती है :

(A) 8 (B) 6 (C) 4 (D) 2

Ans. B

Sol. The slope of the chord is $m = -\frac{8}{k}$ $\Rightarrow$ $k = \pm 1, \pm 2, \pm 4, \pm 8$



but (8, k) must also lie inside the circle $x^2 + y^2 = 125$

$\Rightarrow$ $64 + k^2 - 125 < 0$ $\Rightarrow$ $k^2 < 61$

$\Rightarrow$ k can be equal to $\pm 1, \pm 2, \pm 4$ $\Rightarrow$ 6 values

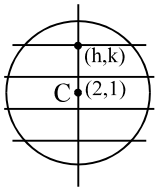
4. Locus of the middle points of a system of parallel chords with slope 2, of the circle $x^2 + y^2 - 4x - 2y - 4 = 0$, has the equation :

उन सभी ढाल 2 के समानान्तर जीवाओं के निकाय के मध्य बिन्दुओं का बिन्दुपथ ज्ञात करें जो वृत $x^2 + y^2 - 4x - 2y - 4 = 0$ के सापेक्ष लिखी गई है :

(A) $x + 2y - 4 = 0$ (B) $x - 2y = 0$ (C) $2x - y - 3 = 0$ (D) $2x + y - 5 = 0$

Ans. A

Sol. Locus will be a line with slope – 1/2
and passing through the centre (2, 1) of the circle



$y - 1 = -\frac{1}{2}(x - 2)$

$2y - 2 = -x + 2$ $\Rightarrow$ $x + 2y - 4 = 0$ Ans.

5. The feet of the perpendicular from the origin on a variable chord of the circle $x^2 + y^2 - 2x - 2y = 0$ is N. If the variable chord makes an angle of 90° at the origin, then the locus of N has the equation :

वृत्त $x^2 + y^2 - 2x - 2y = 0$ की चर जीवा पर मूल बिन्दु से लम्ब का पाद N है। यदि यह चर जीवा मूल बिन्दु पर 90° का कोण बना रही है तो N का बिन्दुपथ होगा :

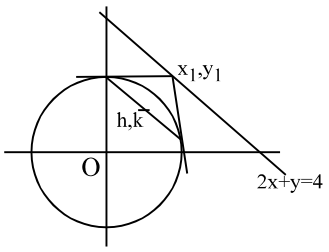
(A) $x^2 + y^2 - x - y = 0$ (B) $x^2 + y^2 + x + y = 0$

(C) $x^2 + y^2 - 2x - 2y = 0$ (D) $x^2 + y^2 + 2x - 2y = 0$

Ans. A

Sol. (x_1, y_1) lies on $2x + y = 4$

$\Rightarrow$ $2x_1 + y_1 = 4$ ....(1)



chord of contact w.r.t. (x_1, y_1)

$xx_1 + yy_1 = 1$

also equation of chord whose mid point is (h, k)

$h^2 + k^2 = hx + ky$

$\therefore$ $\frac{x_1}{h} = \frac{y_1}{k} = \frac{1}{h^2 + k^2}$ $\Rightarrow$ $x_1 = \frac{h}{h^2 + k^2}$; $y_1 = \frac{k}{h^2 + k^2}$

substitute in (1)

$$2 \cdot \frac{h}{h^2+k^2} + \frac{k}{h^2+k^2} = 4$$

locus $= 4(x^2 + y^2) = 2x + y$

6. A circle of constant radius 'a' passes through origin 'O' and cuts the axes of co–ordinates in points P and Q, then the equation of the locus of the foot of perpendicular from O to PQ is :

a त्रिज्या का एक वृत्त O (0, 0)' से जाता है तथा निर्देशांक अक्षों को बिन्दु P तथा Q पर काटता है तो बिन्दु O से रेखा PQ पर डाले गये लम्ब के पाद का बिन्दुपथ होगा :

(A) $(x^2 + y^2)\left(\frac{1}{x^2} + \frac{1}{y^2}\right) = 4a^2$ (B) $(x^2 + y^2)^2\left(\frac{1}{x^2} + \frac{1}{y^2}\right) = a^2$

(C) $(x^2 + y^2)^2\left(\frac{1}{x^2} + \frac{1}{y^2}\right) = 4a^2$ (D) $(x^2 + y^2)\left(\frac{1}{x^2} + \frac{1}{y^2}\right) = a^2$

Ans. C

Sol.

7. The equation of the locus of the mid points of the chords of the circle $4x^2 + 4y^2 - 12x + 4y + 1 = 0$ that subtend an angle of $\frac{2\pi}{3}$ at its centre is :

ऐसी जीवा के मध्य बिन्दु का बिन्दुपथ बताओ जो कि वृत्त $4x^2 + 4y^2 - 12x + 4y + 1 = 0$ के केन्द्र पर $\frac{2\pi}{3}$ कोण बना रहा हो :

(A) $16(x^2 + y^2) - 48x + 16y + 31 = 0$ (B) $16(x^2 + y^2) - 48x - 16y + 31 = 0$

(C) $16(x^2 + y^2) + 48x + 16y + 31 = 0$ (D) $16(x^2 + y^2) + 48x - 16y + 31 = 0$

Ans. A

Sol.

8. The locus of the foot of the perpendicular from the origin upon chords of the circle $x^2 + y^2 - 2x - 4y - 4 = 0$, which subtend a right angle at the origin is :

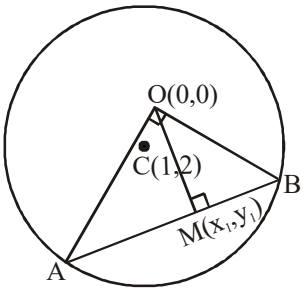
वृत्त $x^2 + y^2 - 2x - 4y - 4 = 0$ की चर जीवाऐं पर मूल बिन्दु से डाले गये लम्ब के पाद का बिन्दु पथ, जहाँ चर जीवा मूल बिन्दु पर 90° को कोण बनाते है, होगा :

(A) $x^2 + y^2 - x - 2y - 2 = 0$ (B) $2(x^2 + y^2) - 2x - 4y + 3 = 0$

(C) $x^2 + y^2 - 2x - 4y + 4 = 0$ (D) $x^2 + y^2 + x + 2y - 2 = 0$

Ans. A

Sol. Equation to the chord AB is $(y-y_1)=\frac{-x_1}{y_1}(x-x_1)$



$\Rightarrow \quad xx_1+yy_1=x_1^2+y_1^2 \quad .....(1)$

Where M (x_1, y_1) is the foot of perpendicular from the origin.

Now homogenising the equation of the given circle, we get

$(x^2+y^2)(x_1^2+y_1^2)^2-(2x+4y)(xx_1+yy_1)(x_1^2+y_1^2)-4(xx_1+yy_1)^2=0$

This represents a pair of perpendicular lines passing through the origin.

Hence coefficient of x^2 + coefficient of $y^2=0$

$\Rightarrow \quad 2(x_1^2+y_1^2)^2-(2x_1(x_1^2+y_1^2)+4y_1(x_1^2+y_1^2))-4(x_1^2+y_1^2)=0$

or $\quad (x_1^2+y_1^2)-(x_1+2y_1)-2=0$

Hence locus of $M(x_1,y_1)$ is $x^2+y^2-x-2y-2=0$ Ans.

9. The locus of the mid points of the chords of the circle $x^2+y^2-ax-by=0$ which subtend a right angle at $(a/2, b/2)$ is :

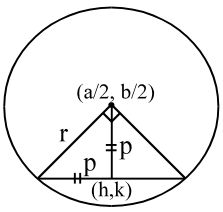
वृत $x^2+y^2-ax-by=0$ के उन जीवाओं के मध्य बिन्दु का बिन्दुपथ बताओं जो कि $(a/2, b/2)$ पर 90° बना रहा है :

(A) $ax+by=0$ (B) $ax+by=a^2+b^2$

(C) $x^2+y^2-ax-by+\frac{a^2+b^2}{8}=0$ (D) $x^2+y^2-ax-by-\frac{a^2+b^2}{8}=0$

Ans. C

Sol. $r=\sqrt{\frac{a^2}{4}+\frac{b^2}{4}}=\frac{\sqrt{a^2+b^2}}{2}; \quad r^2=2p^2$



or $\quad \frac{a^2+b^2}{4}=2\left(\left(x-\frac{a}{2}\right)^2+\left(y-\frac{b}{2}\right)^2\right)$

$\frac{a^2+b^2}{8}=x^2+y^2-ax-by+\frac{a^2+b^2}{4} \quad \Rightarrow$ result

10. The locus of the mid–points of the chords of the circle $x^2+y^2-2x-4y-11=0$ which subtend 60^0 at the centre is :

वृत्त $x^2+y^2-2x-4y-11=0$ की उन जीवाओ के मध्य बिन्दुओं का बिन्दुपथ बताओं जो कि वृत्त केन्द्र पर 60^0 का कोण बनाते हैं:

(A) $x^2+y^2-4x-2y-7=0$ (B) $x^2+y^2+4x+2y-7=0$

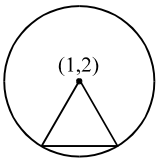
(C) $x^2+y^2-2x-4y-7=0$ (D) $x^2+y^2+2x+4y+7=0$

Ans. C

Sol. (h, k) are mid points of chord

$$\frac{\sqrt{(1-h)^2+(2-k)^2}}{4} = \cos 30°$$

$$= (1-h)^2 + (2-k)^2 = \left(\frac{\sqrt{3}}{2}\times 4\right)^2$$



$h^2 + 1 - 2h + k^2 + 4 - 4k = 12$

hence locus is $x^2 + y^2 - 2x - 4y - 7 = 0$

11. The locus of the middle point of the chord of the circle $x^2 + y^2 = 1$ such that the segment of the chord on the parabola $y = x^2 - x$ subtends a right angle at the origin, is a circle whose centre and radius respectively are :

वृत $x^2 + y^2 = 1$ की एक जीवा के मध्य बिन्दु का बिन्दुपथ एक वृत है तथा इस जीवा का परवलय $y = x^2 - x$ के द्वारा काटा गया अन्तःखण्ड मूल बिन्दु पर 90° कोण बनाता है तो उस वृत का केन्द्र तथा त्रिज्या होगी :

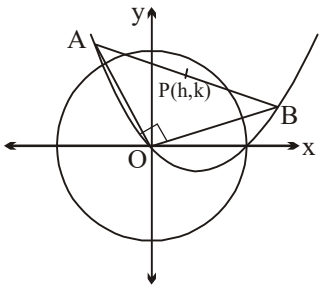
(A) (1, 1) and $\sqrt{2}$ (B) (1, 1) and 2 (C) $\left(\frac{1}{2},\frac{1}{2}\right)$ and $\frac{1}{2}$ (D) $\left(\frac{1}{2},\frac{1}{2}\right)$ and $\frac{1}{\sqrt{2}}$

Ans. D

Sol. Equation of chord of the circle with mid point (h, k) is

$hx + ky = h^2 + k^2$

On homogenising the parabola, we get



$$x^2 - \frac{(x+y)(hx+ky)}{h^2+k^2} = 0$$

$\Rightarrow$ $(h^2 + k^2)x^2 - (x + y)(hx + ky) = 0$

Now coefficient of x^2 + coefficient of $y^2 = 0$

$\therefore$ $h^2 + k^2 - h - k = 0$

$\Rightarrow$ $x^2 + y^2 - x - y = 0 \Rightarrow$ circle with centre $\left(\frac{1}{2},\frac{1}{2}\right)$ and radius $= \sqrt{\frac{1}{4}+\frac{1}{4}} = \frac{1}{\sqrt{2}} \Rightarrow$ (D)

12. The distance between the chords of contact of tangents to the circle $x^2+y^2+2gx+2fy+c=0$ from the origin and the point (g, f) is :

बिन्दु (g, f) से एवं मूल बिन्दु से $x^2+y^2+2gx+2fy+c=0$ पर डाली गयी स्पर्श जीवाओं के मध्य दूरी है :

(A) $\sqrt{g^2+f^2}$ (B) $\frac{\sqrt{g^2+f^2-c}}{2}$ (C) $\frac{g^2+f^2-c}{2\sqrt{g^2+f^2}}$ (D) $\frac{\sqrt{g^2+f^2+c}}{2\sqrt{g^2+f^2}}$

Ans. C

Sol. Equation of chord or contact of the pair of tangent from (0, 0) and (g, f) are

$gx + fy + c = 0$ ....(1) and $gx + fy + \frac{g^2 + f^2 + c}{2} = 0$ ....(2)

These lines are parallel

hence distance $= \left| \frac{c - \frac{g^2 + f^2 + c}{2}}{\sqrt{g^2 + f^2}} \right|$

13. If a circle of constant radius 3k passes through the origin 'O' and meets co-ordinate axes at A and B then the locus of the centroid of the triangle OAB is :

स्थिर त्रिज्या 3k वाला तथा मूलबिन्दु से गुजरने वाला वृत यदि निर्देशांक अक्षो को क्रमशः A एवं B पर काटता है तो त्रिभुज OAB के केन्द्रक का बिन्दुपथ होगा :

(A) $x^2 + y^2 = (2k)^2$ (B) $x^2 + y^2 = (3k)^2$ (C) $x^2 + y^2 = (4k)^2$ (D) $x^2 + y^2 = (6k)^2$

Ans. A

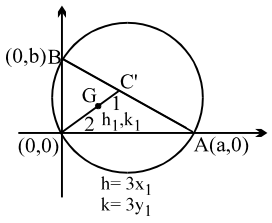
Sol. Given $r = 3k$ and P is centroid

$g^2 + f^2 = (3k)^2$ ....(1) because $c = 0$

$\therefore$ $3h_1 = a$ and $3k_1 = b$ also $2\sqrt{g} = a \Rightarrow 2g = a$ and $2f = b$

$\Rightarrow$ $2g = \frac{3h_1}{2}$ and $2f = \frac{3k_1}{2}$ substitute in (1)

$\frac{9h_1^2}{4} + \frac{9k_1^2}{4} = 9k^2 \quad \Rightarrow \quad x^2 + y^2 = (2k)^2$



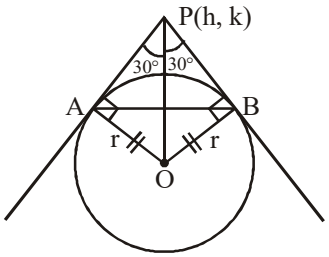
14. Tangents PA and PB are drawn to the circle $x^2 + y^2 = 4$, then the locus of the point P if the triangle PAB is equilateral, is equal to :

वृत $x^2 + y^2 = 4$ पर बिन्दु P से खीचीं गयी स्पर्श रेखा PA एवं PB इस प्रकार से है कि त्रिभुज PAB एक समबाहु त्रिभुज है तो P का बिन्दुपथ होगा :

(A) $x^2 + y^2 = 16$ (B) $x^2 + y^2 = 8$ (C) $x^2 + y^2 = 64$ (D) $x^2 + y^2 = 32$

Ans. A

Sol. Since ΔPAB is equilateral,



$$\Rightarrow \quad \tan 30° = \frac{2}{AP} = \frac{2}{\sqrt{h^2 + k^2 - 4}} = \frac{1}{\sqrt{3}}$$

$$\Rightarrow \quad h^2 + k^2 = 16$$

$\therefore$ Locus of P(h , k) is $x^2 + y^2 = 16$ Ans.

15. Tangents are drawn to a unit circle $x^2 + y^2 = 1$ with centre at the origin from each point on the line $2x + y = 4$. Then the equation to the locus of the middle point of the chord of contact is :

(A) $2(x^2 + y^2) = x + y$ (B) $2(x^2 + y^2) = x + 2y$

(C) $4(x^2 + y^2) = 2x + y$ (D) none

रेखा $2x + y = 4$ के प्रत्येक बिन्दु से वृत $x^2 + y^2 = 1$ पर स्पर्श रेखायें खीचीं जाती है तो उनके स्पर्श जीवा के मध्य बिन्दु का बिन्दुपथ होगा:

(A) $2(x^2 + y^2) = x + y$ (B) $2(x^2 + y^2) = x + 2y$

(C) $4(x^2 + y^2) = 2x + y$ (D) कोई नहीं

Ans. C

Sol. Equation of the circle touching the line $2x + y + 2 = 0$ at $(-1, 1)$ is

$$(x + 1)^2 + (y - 1)^2 + \lambda(3x + y + 2) = 0$$

it passes through $(3, 5)$

$\therefore \quad 16 + 16 + \lambda(9 + 5 + 2) = 0 \quad \Rightarrow \quad \lambda = -2$

$\therefore$ equation of circle is

$$(x + 1)^2 + (y - 1)^2 - 2(3x + y + 2) = 0$$

$$x^2 + y^2 - 4x - 4y - 2 = 0$$

x-coordinate of centre is 2

16. Find the co-ordinates of the middle point of the chord which the circle $x^2 + y^2 + 4x - 2y - 3 = 0$ cuts off the line $x - y + 2 = 0$.

रेखा $x - y + 2 = 0$ द्वारा वृत $x^2 + y^2 + 4x - 2y - 3 = 0$ पर काटी गयी जीवा के मध्य बिन्दु के निर्देशांक ज्ञात करें।

Ans. $\left(\frac{-3}{2}, \frac{1}{2}\right)$

Sol. Centre and radius of the circle $x^2 + y^2 + 4x - 2y - 3 = 0$ are $(-2, 1)$ and $\sqrt{4+1+3} = 2\sqrt{2}$ repectively.

Draw perpendicular from O upon $x - y + 2 = 0$ is OM.

Equation of OM which is perpendicular to $x - y + 2 = 0$ is $x + y = \lambda$, it passes through $(2, 1)$

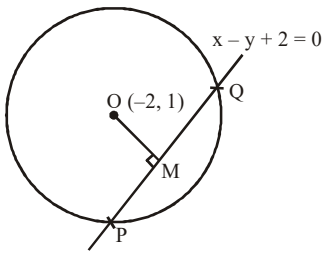
Then $-2 + 1 = \lambda$

$\therefore \quad \lambda = -1$

then equation of OM is $x + y + 1 = 0$

Since M is the mid point of PQ which is point of intesection of $x - y + 2 = 0$ and $x + y + 1 = 0$,

co-ordinates of M is $\left(\frac{-3}{2}, \frac{1}{2}\right)$



17. Show that the point $(1, 2)$ lies inside the circle $x^2 + y^2 - 11x - 2y - 24 = 0$ and find the equation of the chord of which it is the middle point.

दर्शाइये कि बिन्दु $(1, 2)$ वृत $x^2 + y^2 - 11x - 2y - 24 = 0$ के अन्दर स्थित है तथा इस बिन्दु को मध्य बिन्दु मानते हुये जीवा का समीकरण लिखियें।

Ans. $9x - 2y - 5 = 0$

Sol.

18. Find the co-ordinates of the middle point of the chord which the circle $x^2 + y^2 - 2x + 2y - 2 = 0$ cuts off on the line $y = x - 1$. Find also the equation of the locus of the middle point of all chords of the circle which are parallel to the line $y = x - 1$.

एक रेखा $y = x - 1$ द्वारा वृत $x^2 + y^2 - 2x + 2y - 2 = 0$ पर काटी गयी जीवा का मध्य बिन्दु ज्ञात करें। साथ ही उन सभी जीवाओं के मध्य बिन्दु का बिन्दुपथ ज्ञात करें जो रेखा $y = x - 1$ के समानान्तर है।

Ans. $\left(\frac{1}{2}, -\frac{1}{2}\right)$, $x + y = 0$

Sol.

19. A chord of the circle $x^2 - 2ax + y^2 = 0$ is drawn to pass through the origin. Show that the locus of the centre of the circle described on this chord as diameter is a circle, passing through the centre of the given circle.

वृत $x^2 - 2ax + y^2 = 0$ की एक जीवा मूल बिन्दु से होकर जाती है। सिद्ध कीजिये कि जीवा को व्यास मानकर बनाए वत के केन्द्र का बिन्दुपथ दिये गये वृत के केन्द्र से होकर गुजरता है।

Ans.

Sol.

20. The Straight line $x-2y+1=0$ intersects the circle $x^2+y^2=25$ in points T and T', find the co-ordinates of a point of intersection of tangents drawn at T and T' to the circle.

सरल रेखा $x - 2y + 1 = 0$ वृत $x^2 + y^2 = 25$ को बिन्दुओं T तथा T' पर काटती है। बिन्दुओं T तथा T' से वृत पर स्पर्श रेखाओं का प्रतिच्छेदन बिन्दु ज्ञात कीजिए।

Ans. $(-25, 50)$

Sol.

1. Sum of the abscissa and ordinate of the centre of the circle touching the line $3x + y + 2 = 0$ at the point $(-1, 1)$ and passing through the point $(3, 5)$ is :

बिन्दु $(3, 5)$ से गुजरने वाली तथा रेखा $3x + y + 2 = 0$ को $(-1, 1)$ पर स्पर्श करने वाली वृत के केन्द्र की भुज एवं कोटि का योग होगा:

(A) 2 (B) 3 (C) 4 (D) 5

Ans. C

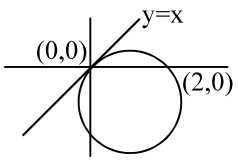
Sol. $x^2 + y^2 + \lambda(y - x) = 0$

$4 + \lambda(0 - 2) = 0$

$4 - 2\lambda = 0$

$\lambda = 2$

$x^2 + y^2 + 2y - 2x = 0$ Ans.



2. A circle touches the bisector of the first and third quadrant at the origin and passes through the point $(2, 0)$. The equation of the circle is :

(A) $x^2 + y^2 - 2x - 2y = 0$ (B) $x^2 + y^2 - 2x + 2y = 0$

(C) $x^2 + y^2 + 2x + 2y = 0$ (D) none of these

एक वृत प्रथम एवं तृतीय चतुर्थांश के कोणार्धक को मूल बिन्दु पर स्पर्श करता है तथा बिन्दु $(2, 0)$ से गुजरता है, तो वृत का समीकरण होगा:

(A) $x^2 + y^2 - 2x - 2y = 0$ (B) $x^2 + y^2 - 2x + 2y = 0$

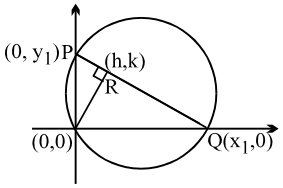
(C) $x^2 + y^2 + 2x + 2y = 0$ (D) इनमें से कोई नहीं

Ans. B

Sol. Equation of line PQ is

$$y - k = -\frac{h}{k}(x - h) \quad \Rightarrow \quad hx + ky = h^2 + k^2 \text{ ; also } 2a = \sqrt{x_1^2 + y_1^2} \Rightarrow x_1^2 + y_1^2 = 4a^2$$

$\therefore$ points $Q\left(\frac{h^2 + k^2}{h}, 0\right)$ and $P\left(0, \frac{h^2 + k^2}{k}\right)$



$$(x^2 + y^2)^2\left(\frac{1}{x^2} + \frac{1}{y^2}\right) = 4a^2$$

3. A circle of radius 5 is tangent to the line $4x - 3y = 18$ at $M(3, -2)$ and lies above the line. The equation of the circle, is :

5 त्रिज्या वाले तथा रेखा $4x - 3y = 18$ को $M(3, -2)$ पर स्पर्श करने वाली तथा उपरोक्त रेखा के उपर स्थित वृत्त होगा :

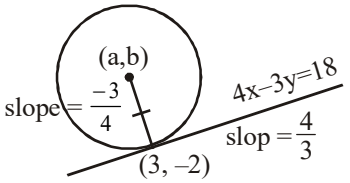
(A) $x^2 + y^2 - 6x + 4y - 12 = 0$ (B) $x^2 + y^2 + 2x - 2y - 3 = 0$

(C) $x^2 + y^2 + 2x - 2y - 23 = 0$ (D) $x^2 + y^2 + 6x + 4y - 12 = 0$

Ans. C

Sol. Let $\frac{x-3}{\cos\theta} = \frac{y+2}{\sin\theta} = 5$ where $\tan\theta = \frac{-3}{4}$

$$\left.\begin{aligned} x &= 5\cos\theta + 3 \\ y &= 5\sin\theta - 2 \end{aligned}\right]$$



$\Rightarrow \cos\theta = \frac{-4}{5}$; $\sin\theta = \frac{3}{5}$

$\therefore$ $x = -1$ and $y = 1$ $\Rightarrow$ centre $(-1, 1)$

Hence required equation of circle is $(x + 1)^2 + (y - 1)^2 = 25 \Rightarrow x^2 + y^2 + 2x - 2y - 23 = 0$ Ans.

4. Two circles of radii 4 cms and 1 cm touch each other externally and θ is the angle contained by their direct common tangents. Then $\sin\theta =$

(A) $\frac{24}{25}$ (B) $\frac{12}{25}$ (C) $\frac{3}{4}$ (D) none

दो वृत जिनकी त्रिज्यायें 4 cms तथा 1 cm है एक दूसरे को बाह्य स्पर्श करते है तथा θ उनकी उभयनिष्ठ अनुस्पर्शी रेखाओं के मध्य कोण है। तो $\sin\theta =$

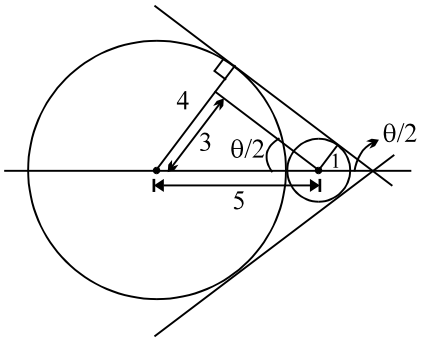
(A) $\frac{24}{25}$ (B) $\frac{12}{25}$ (C) $\frac{3}{4}$ (D) कोई नही

Ans. A

Sol. $\sin\frac{\theta}{2} = \frac{3}{5}$

$\cos\frac{\theta}{2} = \frac{4}{5}$

$\therefore$ $\sin\theta = 2 \cdot \frac{3}{5} \cdot \frac{4}{5} = \frac{24}{25}$ Ans.



5. The locus of the centre of a circle which touches externally the circle, $x^2 + y^2 - 6x - 6y + 14 = 0$ and also touches the y-axis is given by the equation :

ऐसे वृत्त के केन्द्र का बिन्दु पथ बताओ जो कि वृत्त $x^2 + y^2 - 6x - 6y + 14 = 0$ को बाह्य स्पर्श करता है तथा y-अक्ष को भी स्पर्श

करता है :

(A) $x^2 - 6x - 10y + 14 = 0$ (B) $x^2 - 10x - 6y + 14 = 0$

(C) $y^2 - 6x - 10y + 14 = 0$ (D) $y^2 - 10x - 6y + 14 = 0$

Ans. D

Sol.

6. The circle $x^2 + y^2 - 4x + 4y = 0$ with centre C_1 and $x^2 + y^2 + 2x - 2y = 0$ with centre C_2 touch each other at point P. Then the coordinates of point Q representing the harmonic conjugate of point P with respect to C_1 and C_2 are :

(A) $(-4, 4)$ (B) $(4, -4)$ (C) $(4, 4)$ (D) None

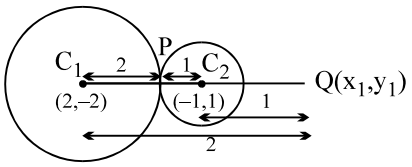
वृत्त $x^2 + y^2 - 4x + 4y = 0$ तथा $x^2 + y^2 + 2x - 2y = 0$ का केन्द्र क्रमशः C_1 एवं C_2 है और जो कि एक दूसरे को बिन्दु P पर स्पर्श करते है। यदि C_1 एवं C_2 के सापेक्ष बिन्दु P का हरात्मक सयुग्मी यदि Q है तो Q का निर्देशांक होगा :

(A) $(-4, 4)$ (B) $(4, -4)$ (C) $(4, 4)$ (D) कोई नही

Ans. A

Sol. radius of circle with centre $C_1 = 2\sqrt{2}$

and radius of circle with centre $C_2 = \sqrt{2}$



$x_1 + 2 = -2$ $y_1 - 2 = 2$

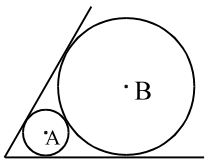
$\therefore \quad x_1 = -4$ $\therefore \quad y_1 = 4$

Q is the harmonic conjugate pf P w.r.t. C_1 and C_2

$\therefore \quad C_1Q : C_2Q = 2 : 1$

7. A circle with center A and radius 7 is tangent to the sides of an angle of 60°. A larger circle with center B is tangent to the sides of the angle and to the first circle. The radius of the larger circle is :

दिये गये चित्रनुसार यदि छोटी वृत्त की त्रिज्या 7 है तथा स्पर्श रेखाओं के बीच कोण 60° है तो बड़ी वृत्त की त्रिज्या होगी :



(A) $30\sqrt{3}$ (B) 21

(C) $20\sqrt{3}$ (D) 30

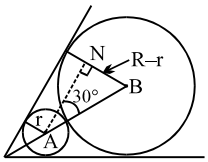
Ans. B

Sol. r = 7

$$\sin 30° = \frac{R-r}{R+r} = \frac{1}{2}$$

$2R - 2r = R + r$

$R = 3r = 21$ Ans.



8. The number of common tangents of the circles $(x+2)^2 + (y-2)^2 = 49$ and $(x-2)^2 + (y+1)^2 = 4$ is :

वृत्त $(x+2)^2 + (y-2)^2 = 49$ तथा $(x-2)^2 + (y+1)^2 = 4$ की कुल उभयनिष्ठ स्पर्श रेखाओ की संख्या होगी :

(A) 0 (B) 1 (C) 2 (D) 3

Ans. B

Sol.

9. Let C_1 and C_2 are circles defined by $x^2 + y^2 - 20x + 64 = 0$ and $x^2 + y^2 + 30x + 144 = 0$. The length of the shortest line segment PQ that is tangent to C_1 at P and to C_2 at Q is :

यदि $C_1 : x^2 + y^2 - 20x + 64 = 0$ तथा $C_2 : x^2 + y^2 + 30x + 144 = 0$ तो सबसे छोटी उभयनिष्ठ स्पर्श रेखा PQ की लम्बाई होगी :

(A) 15 (B) 18 (C) 20 (D) 24

Ans. C

Sol. Centres are (10, 0) and (– 15, 0)

$r_1 = 6$; $r_2 = 9$

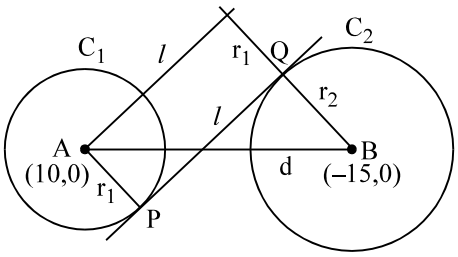
$d = 25$

$r_1 + r_2 < d$

$\Rightarrow$ circles are separated

$PQ = l = \sqrt{d^2 - (r_1 + r_2)^2}$

$= \sqrt{625 - 225} = 20$



10. If the line $x\cos\theta + y\sin\theta = 2$ is the equation of a transverse common tangent to the circles $x^2 + y^2 = 4$ and $x^2 + y^2 - 6\sqrt{3}\,x - 6y + 20 = 0$, then the value of θ is :

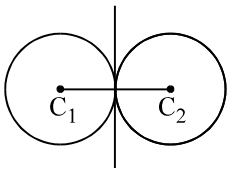
यदि $x\cos\theta + y\sin\theta = 2$ वृत्त $x^2 + y^2 = 4$ तथा $x^2 + y^2 - 6\sqrt{3}\,x - 6y + 20 = 0$ की उभयनिष्ठ तिर्यक स्पर्श रेखा है तो θ का मान होगा :

(A) $5\pi/6$ (B) $2\pi/3$ (C) $\pi/3$ (D) $\pi/6$

Ans. D

Sol. $C_1C_2 = r_1 + r_2$

$C_1 = (0, 0)$; $C_2 = (3\sqrt{3}, 3)$ and $r_1 = 2, r_2 = 4$



$\Rightarrow$ circle touch each other externally

equation of common tangent is, $S_1 - S_2 = 0$ i.e. $\sqrt{3}\,x + y - 4 = 0$ ....(1)

comparing it with $x\cos\theta + y\sin\theta = 2$

$$\theta = \frac{\pi}{6}$$

11. Two circles have an external tangent with length 36 cm. The shortest distance between these circles is 14 cm. If the radius of the longer circle is 4 times the radius of the smaller circle then the radius of the larger circle in cms is :

दो वृत्तों की बाह्य स्पर्श रेखा की लम्बाई 36 cm है तथा वृतों के मध्य न्यूनतम दूरी 14 cm है। यदि बड़े वृत्त की त्रिज्या, छोटे वृत्त की त्रिज्या की चार गुनी है तो बड़े वृत्त की त्रिज्या सेमी में ज्ञात कीजिये :

(A) 12 (B) 20 (C) 24 (D) 32

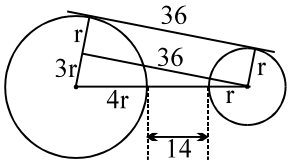
Ans. B

Sol. $9r^2 + (36)^2 = (5r + 14)^2$

$\Rightarrow$ $4r^2 + 35r - 25 \cdot 11 = 0$ $\Rightarrow$ $(r - 5)(4r + 55) = 0$

hence $r = 5$

radius of the larger circle is 20



Paragraph for Question nos. 12 to 14

Consider two circles $S_1 = 0$ and $S_2 = 0$, each of radius 1unit touching internally the sides of ΔOAB and ΔABC respectively where $O \equiv (0, 0)$, $A \equiv (0, 4)$ and B, C are the points on positive x-axis such that $OB < OC$.

इकाई त्रिज्या वाले दो वृत $S_1 = 0$ एवं $S_2 = 0$, ΔOAB तथा ΔABC को आन्तरिक रूप से स्पर्श करते है जहाँ $O \equiv (0, 0)$, $A \equiv (0, 4)$ तथा B, C धनात्मक x-अक्ष पर इस प्रकार से बिन्दु है कि $OB < OC$

12. If the angle subtended by the circle $S_1 = 0$ at point A is θ, then $\cos\theta$ equals :

यदि वृत $S_1 = 0$ बिन्दु A पर θ कोण बनाये तो $\cos\theta$ का मान होगा :

(A) $\frac{3}{5}$ (B) $\frac{2}{3}$ (C) $\frac{3}{4}$ (D) $\frac{4}{5}$

Ans. D

Sol.

13. The length of tangent from A to the circle $S_2 = 0$, is :

बिन्दु A से वृत $S_2 = 0$ पर खीचीं गयी स्पर्श रेखा की लम्बाई होगी :

(A) $\frac{5}{2}$ (B) $\frac{7}{2}$ (C) $\frac{9}{2}$ (D) $\frac{11}{2}$

Ans. C

Sol.

14. If one of the diameter of the circle $S_2 = 0$ is a chord to the circle $S_3 = 0$ with centre $\left(\frac{3}{2}, 3\right)$, then radius of circle $S_3 = 0$, is :

यदि $S_2 = 0$ की व्यास वृत $S_3 = 0$ की एक जीवा है जिसका केन्द्र $\left(\frac{3}{2}, 3\right)$ है तो वृत $S_3 = 0$ की त्रिज्या होगी :

(A) $\sqrt{3}$ (B) 3 (C) $\sqrt{2}$ (D) 2

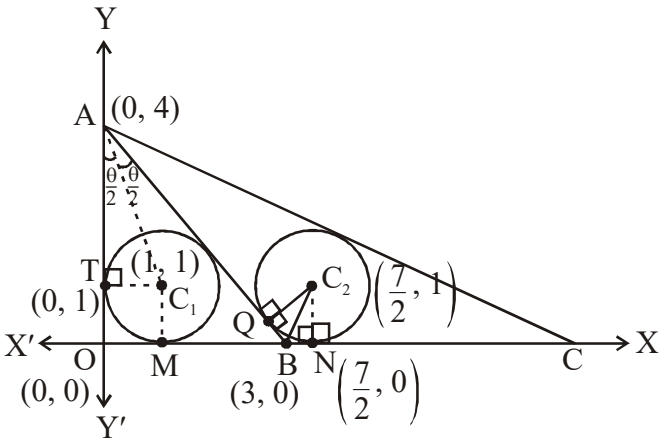
Ans. B

Sol. (i) We have centre of circle $S_1 = 0$ is $C_1 \equiv (1, 1)$

Now in ΔATC_1, $\tan\frac{\theta}{2} = \frac{1}{3}$

$$\therefore \tan\theta = \frac{2\tan\frac{\theta}{2}}{1-\tan^2\frac{\theta}{2}} = \frac{2\times\frac{1}{3}}{1-\left(\frac{1}{3}\right)^2} = \frac{\frac{2}{3}}{\frac{8}{9}} = \frac{3}{4}$$

Hence $\cos\theta = \frac{4}{5}$



(ii) We have $\angle ABC = \frac{\pi}{2} + \theta$

$$\therefore \quad \angle QBC_2 = \frac{1}{2}\angle ABC = \left(\frac{\pi}{4} + \frac{\theta}{2}\right)$$

Now in ΔQBC_2, $\frac{QC_2}{QB} = \tan\left(\frac{\pi}{4} + \frac{\theta}{2}\right) \Rightarrow QB = \frac{1}{\tan\left(\frac{\pi}{4} + \frac{\theta}{2}\right)}$

$$\therefore \quad QB = \frac{1-\tan\frac{\theta}{2}}{1+\tan\frac{\theta}{2}} = \frac{1-\frac{1}{3}}{1+\frac{1}{3}} = \frac{\frac{2}{3}}{\frac{4}{3}} = \frac{1}{2}$$

Also $\tan\theta = \frac{3}{4} = \frac{OB}{OA}$

$\Rightarrow \quad OB = \frac{3}{4}(OA) = \frac{3}{4} \times 4 = 3$

$\therefore \quad B = (3, 0)$

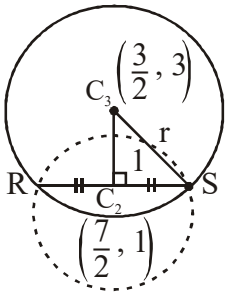
Hence length of tangent from A to circle $S_2 = 0$ is $AQ = AB - QB = 5 - \frac{1}{2} = \frac{9}{2}$

($\because$ AB = distance between A(0, 4) and B(3, 0) = 5)

(iii) We know that $BQ = BN = \frac{1}{2}$,

So $N = \left(\frac{7}{2}, 0\right)$

$\therefore$ Centre of circle $S_2 = 0$ is $C_2 = \left(\frac{7}{2}, 1\right)$



Let r = radius of circle $S_3 = 0$, So in ΔC_3C_2S,

$(C_2S)^2 = (C_3C_2)^2 + (C_2S)^2$

$\Rightarrow \quad r^2 = \left(\frac{7}{2} - \frac{3}{2}\right)^2 + (3-1)^2 + 1 = 4 + 4 + 1$

Hence r = 3

15. Find the equation of the circle passing through the point of intersection of the circles $x^2 + y^2 - 6x + 2y + 4 = 0$, $x^2 + y^2 + 2x - 4y - 6 = 0$ and with its centre on the line y = x :

वृतों $x^2 + y^2 - 6x + 2y + 4 = 0$ एवं $x^2 + y^2 + 2x - 4y - 6 = 0$ के प्रतिच्छेदन बिन्दुओं से गुजरने वाले वृत का समीकरण ज्ञात करें जिसका केन्द्र y = x पर स्थित है :

Ans. $7x^2 + 7y^2 - 10x - 10y - 12 = 0$

Sol. Equation of any circle through the points of intersection of given circles is

$(x^2 + y^2 - 6x + 2y + 4) + \lambda (x^2 + y^2 + 2x - 4y - 6) = 0$

$\Rightarrow x^2(1 + \lambda) + y^2(1 + \lambda) - 2x(3 - \lambda) + 2y(1 - 2\lambda) + (4 - 6\lambda) = 0$

or $x^2 + y^2 - \frac{2x(3-\lambda)}{(1+\lambda)} + \frac{2y(1-2\lambda)}{(1+\lambda)} + \frac{(4-6\lambda)}{(1+\lambda)} = 0$ .....(1)

Its centre $\left\{\frac{3-\lambda}{1+\lambda}, \frac{2\lambda-1}{1+\lambda}\right\}$ lies on the line y = x

then $\frac{2\lambda-1}{1+\lambda}=\frac{3-\lambda}{1+\lambda} \Rightarrow \lambda \neq 1$

$\therefore \quad 2\lambda - 1 = 3 - \lambda$

or $3\lambda = 4$

$\therefore \ \lambda = \frac{4}{3}$

$\therefore$ Substituting the value of $\lambda = \frac{4}{3}$ in (1), we get the required equation is

$7x^2 + 7y^2 - 10x + 10y - 12 = 0$ Ans.

16. Find the equation of the circle through points of intersection of the circle $x^2 + y^2 - 2x - 4y + 4 = 0$ and the line $x + 2y = 4$ which touches the line $x + 2y = 0$.

वृतों $x^2 + y^2 - 2x - 4y + 4 = 0$, $x + 2y = 4$ एवं के प्रतिच्छेदन बिन्दुओं से गुजरने वाले वृत का समीकरण ज्ञात करें जो रेखा $x + 2y = 0$ को स्पर्श करता है।

Ans. $x^2 + y^2 - x - 2y = 0$

Sol. Equation of any circle through points of intersection of the given circle and the line is

$(x^2 + y^2 - 2x - 4y - 4) + \lambda(x + 2y - 4) = 0$

or $x^2 + y^2 + (\lambda - 2)x + (2\lambda - 4)y + 4(1 - \lambda) = 0$

It will touch the line $x + 2y = 0$ if solution of equation (1) and $x = -2y$ be unique.

Hence the roots of the equation

$(-2y)^2 + y^2 + (\lambda - 2)(-2y) + (2\lambda - 4)y + 4(1 - \lambda) = 0$

or $5y^2 + 4(1 - \lambda) = 0$

must be equal.

Then $0 - 4 \cdot 5 \cdot 4 (1 - \lambda) = 0$ or $1 - \lambda = 0$ or $\lambda = 1$

From (1), the required circle is $x^2 + y^2 - x - 2y = 0$ Ans.

17. Find the equation of the circle drawn on the common chord of the following pair of circles as diameter $x^2 + y^2 - 6x - 4y = 12$; $x^2 + y^2 - 2x - 6y = 15$.

वृत्तो $x^2 + y^2 - 6x - 4y = 12$ एवं $x^2 + y^2 - 2x - 6y = 15$ की उभयनिष्ठ जीवा को व्यास मानकर बनाये वृत का समीकरण ज्ञात कीजिए।

Ans. $2x^2 + 2y^2 - 8x - 10y - 27 = 0$

Sol.

18. Find the equations of the circles which pass through the common points of the following pair of circles :

(a) $x^2 + y^2 + 2x + 3y - 7 = 0$ and $x^2 + y^2 + 3x - 2y - 1 = 0$ through the point (1, 2).

(b) $x^2 + y^2 + 4x - 6y - 12 = 0$ and $x^2 + y^2 - 5x + 17y = 19$ and having its centre on $x + y = 0$.

उन वृतों का समीकरण ज्ञात करें जो निम्न वृतों के उभयनिष्ठ बिन्दुओं से गुजरते है :

(a) $x^2 + y^2 + 2x + 3y - 7 = 0$ एवं $x^2 + y^2 + 3x - 2y - 1 = 0$ एवं बिन्दु (1, 2) से गुजरने वाला

(b) $x^2 + y^2 + 4x - 6y - 12 = 0$ एवं $x^2 + y^2 - 5x + 17y = 19$ एवं इसके केन्द्र $x + y = 0$ पर है

Ans. (a) $x^2 + y^2 + 4x - 7y + 5 = 0$ (b) $2(x^2 + y^2) - x + y - 31 = 0$

Sol.

19. Find the equation of the circle which touches the circle $x^2 + y^2 - 6x + 6y + 17 = 0$ externally and to which the lines $x^2 - 3xy - 3x + 9y = 0$ are normals.

उस वृत्त का समीकरण बताओ जो कि वृत्त $x^2 + y^2 - 6x + 6y + 17 = 0$ को बाह्य स्पर्श करती है तथा रेखा युग्म $x^2 - 3xy - 3x + 9y = 0$ इसके अभिलम्ब है।

Ans. $x^2 + y^2 - 6x - 2y + 1 = 0$

Sol. The given circle is

$$x^2 + y^2 - 6x + 6y + 17 = 0$$

Centre and radius of the circles are (3, –3) and $\sqrt{9+9-17} = 1$ respectively.

But given the required circle has normals

$x^2 - 3xy - 3x + 9y = 0$

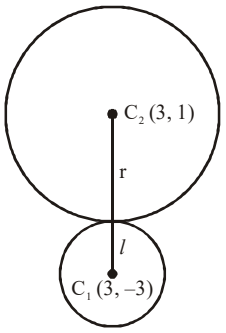
or $(x - 3)(x - 3y) = 0$

or $x = 3$ and $x - 3y = 0$ ....(1)

but point of intersection of normals represented by (1) is (3, 1) which is centre of the required circle. Since given circle and required circle touch each other externally,

then (if radius of required circles is r)

Sum of radii = Distance between the centres



$r + 1 = \sqrt{(3-3)^2 + (-3-1)^2} = 4$

$\therefore \quad r = 3$

$\therefore$ Equation of required circle is $(x - 3)^2 + (y - 1)^2 = 3^2$

or $x^2 + y^2 - 6x - 2y + 1 = 0$ Ans.

1. Circle C_1 has radius 2 and Circle C_2 has radius 3, and the distance between the centers of C_1 and C_2 is 7. If two lines, one tangent to both circles and the other passing through the center of both circles, intersect at at point P which lies between the centers of C_1 and C_2, then the distance between P and the center of C_1 is :

C_1 की त्रिज्या 2, वृत्त C_2 की त्रिज्या 3 एवं उनके केन्द्र के बीच की दूरी 7 है यदि उनकी उभयनिष्ठ स्पर्श रेखा उनके केन्द्र को मिलाने वाली रेखा को यदि बिन्दु P पर काट रही हो, जो कि दोनों वृत्तों के बीच में है तो P C_1 की दूरी बताओ :

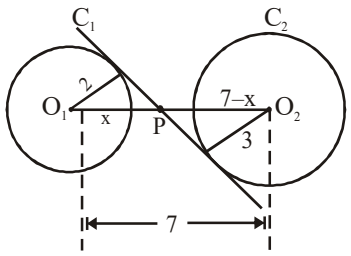
(A) $\frac{9}{4}$ (B) $\frac{7}{3}$ (C) $\frac{8}{3}$ (D) $\frac{14}{5}$

Ans. D

Sol. $\frac{2}{3} = \frac{x}{7-x}$

$3x = 14 - 2x$

$x = \frac{14}{5}$ Ans.



2. If the two circles $(x-1)^2+(y-3)^2=r^2$ and $x^2+y^2-8x+2y+8=0$ intersect in two distinct points then :

यदि वृत्त $(x-1)^2+(y-3)^2=r^2$ तथा $x^2+y^2-8x+2y+8=0$ एक दूसरे को दो भिन्न–भिन्न बिन्दुओं पर काटते है तो :

(A) $2 < r < 8$ (B) $r < 2$ (C) $r = 2$ (4) $r > 2$

Ans. A

Sol.

3. If the circles $x^2+y^2+2ax+cy+a=0$ and $x^2+y^2-3ax+dy-1=0$ intersect in two distinct points P and Q then the line $5x+by-a=0$ passes through P and Q for :

(A) exactly one value of a (B) no value of a

(C) infinitely many values of a (D) exactly two values of a

यदि वृत्त $x^2+y^2+2ax+cy+a=0$ तथा $x^2+y^2-3ax+dy-1=0$ एक दूसरे को P एवं Q पर काटते है यदि रेखा $5x+by-a=0$, P एवं Q से गुजरती है, तो ऐसा सिर्फ :

(A) a के केवल एक मान के लिये होगा (B) a के किसी भी मान के लिये नही होगा

(C) a के अनन्त मान के लिये होगा (D) a के केवल दो मान के लिये होगा

Ans. B

Sol. $S_1 - S_2 = 5ax + (c-d)y + a + 1 = 0$ and $5x + by - a = 0$ must represent the same line

$$\frac{a}{1} = \frac{c-d}{b} = \frac{a+1}{-a}$$

$ab = c - d$ and $+ a^2 + a + 1 = 0$

a is imaginary only

so no value of a $\Rightarrow$ (B)

4. If the circle $C_1 : x^2 + y^2 = 16$ intersects another circle C_2 of radius 5 in such a manner that the common chord is of maximum length and has a slope equal to 3/4, then the co-ordinates of the centre of C_2 are :

यदि वृत्त $C_1 : x^2 + y^2 = 16$ एक दूसरे वृत्त C_2 को, जिसकी त्रिज्या 5 है, पर इस प्रकार से काटती है कि उनकी उभयनिष्ठ जीवा की लम्बाई अधिकतम है तथा उसकी प्रवणता 3/4 है तो C_2 के केन्द्र का निर्देशांक होगा।

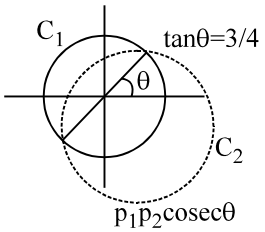
(A) $\left(\pm\frac{9}{5}, \pm\frac{12}{5}\right)$ (B) $\left(\pm\frac{9}{5}, \mp\frac{12}{5}\right)$ (C) $\left(\pm\frac{12}{5}, \pm\frac{9}{5}\right)$ (D) $\left(\pm\frac{12}{5}, \mp\frac{9}{5}\right)$

Ans. B

Sol. Equation of common chord is $y = \frac{3}{4}x \Rightarrow 3x - 4y = 0$

family of circle $x^2 + y^2 - 4 + \lambda(3x - 4y) = 0$

equate radius of this circle as 5. Alternately .....



5. Suppose that two circles C_1 and C_2 in a plane have no points in common. Then :

(A) there is no line tangent to both C_1 and C_2.

(B) there are exactly four lines tangent to both C_1 and C_2.

(C) there are no lines tangent to both C_1 and C_2 or there are exactly two lines tangent to both C_1 and C_2.

(D) there are no lines tangent to both C_1 and C_2 or there are exactly four lines tangent to both C_1 and C_2.

यदि C_1 एवं C_2 वृत्त का कोई बिन्दु उभयनिष्ठ नहीं है, तो :

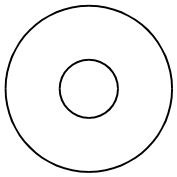
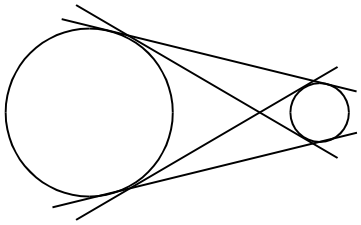
(A) C_1 एवं C_2 के बीच कोई स्पर्श रेखा नही होगी

(B) C_1 एवं C_2 के बीच कोई चार स्पर्श रेखा होगी

(C) या तो 2 अथवा 4 उभयनिष्ठ स्पर्शरेखा होगी

(D) या तो 0 अथवा 4 उभयनिष्ठ स्पर्शरेखा होगी

Ans. D

Sol.  or 

6. Circle K is inscribed in the first quadrant touching the circle $x^2 + y^2 = 36$ internally. The length of the radius of the circle K, is :

प्रथम चतुर्थांश में परिबद्ध तथा वृत्त $x^2 + y^2 = 36$ को आन्तरिक स्पर्श करने वाला एक वृत्त K है तो K की त्रिज्या होगी :

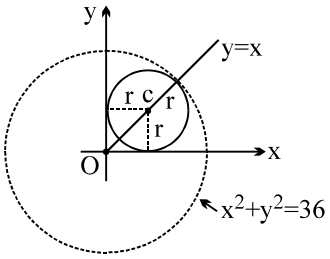
(A) $\frac{6-\sqrt{2}}{2}$ (B) $\frac{3\sqrt{2}}{2}$ (C) 3 (D) $6\left(\sqrt{2}-1\right)$

Ans. D

Sol. $r^2 + r^2 = (OC)^2 \quad \Rightarrow \quad OC = r\sqrt{2}$

$\therefore \quad r\sqrt{2} + r = 6$

$r = \frac{6}{\sqrt{2}+1} = 6\left(\sqrt{2}-1\right)$ Ans.



7. The number of common tangent(s) to the circles $x^2+y^2+2x+8y-23=0$ and $x^2+y^2-4x-10y+19=0$ is :

वृत $x^2+y^2+2x+8y-23=0$ तथा $x^2+y^2-4x-10y+19=0$ की उभयनिष्ठ स्पर्श रेखा की संख्या होगी :

(A) 1 (B) 2 (C) 3 (D) 4

Ans. C

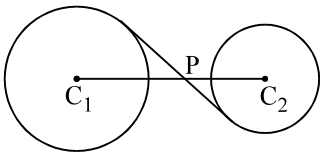
Sol. $C_1 = (-1, -4)$; $C_2 = (2, 5)$; $r_1 = \sqrt{1+16+23} = 2\sqrt{10}$; $r_2 = \sqrt{4+25-19} = \sqrt{10}$

$C_1C_2 = \sqrt{9+81} = 3\sqrt{10} \quad \Rightarrow r_1 + r_2 = C_1C_2 \quad \Rightarrow \quad$ (C)

8. In the figure given, two circles with centres C_1 and C_2 are 35 units apart, i.e. $C_1C_2 = 35$. The radii of the circles with centres C_1 and C_2 are 12 and 9 respectively. If P is the intersection of C_1C_2 and a common internal tangent to the circles, then $l(C_1P)$ equals :

दिखाऐ गये चित्रनुसार केन्द्र C_1 तथा C_2 , 35 मात्रक की दूरी पर है तथा $r_1 = 12$ एवं $r_2 = 9$ है यदि C_1एवं C_2 को मिलाने वाली रेखा

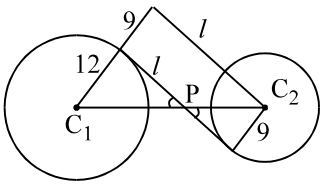
आन्तरिक उभयनिष्ठ स्पर्श रेखा को P पर काटती है तो $l(C_1P)$ का मान होगा :



(A) 18 (B) 20 (C) 12 (D) 15

Ans. B

Sol. Let $C_1P = a$



$PC_2 = 35 - a$

Hence, $\frac{9}{35-a} = \frac{12}{a} \Rightarrow 3a = 4(35-a)$

$\Rightarrow 7a = 4 \cdot 35 \Rightarrow a = 20$ **Ans.**

9. Let C_1 be the circle of radius $r > 0$ with centre at (0, 0) and let C_2 be the circle of radius 'r' with centre at (r, 0). The length of the arc of the circle C_1 that lies inside the circle C_2, is :

माना कि वृत C_1 का केन्द्र (0, 0) तथा त्रिज्या $r > 0$ है एवं एक दूसरे वृत C_2 का केन्द्र (r, 0) तथा त्रिज्या r है तो C_1 की चाप की लम्बाई बताओं जो कि वृत C_2 के भीतर है :

(A) $\frac{\pi r}{3}$ (B) $\frac{2\pi r}{3}$ (C) $\frac{3\pi r}{4}$ (D) $\frac{5\pi r}{6}$

Ans. B

Sol. $C_1 : x^2 + y^2 = r^2$; $C_2 : (x-r)^2 + y^2 = r^2$

solving for x

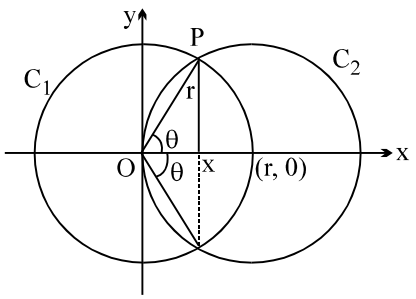
$(x-r)^2 + r^2 - x^2 = r^2$

$x - r = x$ or $-x$

$\therefore x = \frac{r}{2}$

$\therefore r\cos\theta = \frac{r}{2}$

$\theta = \frac{\pi}{3}$



$2\theta = \frac{2\pi}{3}$; length of C_1 inside $C_2 = r(2\theta) = r \cdot \frac{2\pi}{3} = \frac{2\pi r}{3}$

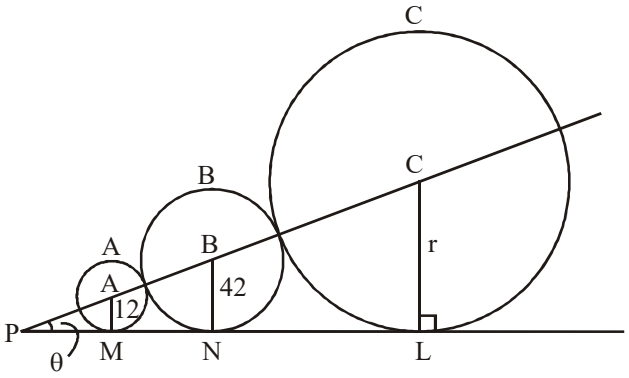
10. The centres of the three circles A, B and C are collinear with the centre of circle B lying between the centres of circle A and C. Circles A and C are both externally tangent to circle B, and the three circles share a common tangent line. Given the circle A has radius 12 and circle B has radius 42. The radius of circle C, is equal to :

तीन वृत्तो A, B तथा C के केन्द्र संरेखीय है जहाँ B , A एवं C के मध्य स्थित है वृत B, A एवं C को बाह्य स्पर्श करता है तथा तीनों वृतों के लिये भी एक उभयनिष्ठ स्पर्श रेखा है यदि वृत A एवं B की त्रिज्या क्रमशः 12 एवं 42 है तो वृत C की त्रिज्या होगी :

(A) 120 (B) 136 (C) 147 (D) 171

Ans. C

Sol. We have $\sin\theta = \frac{12}{PA} = \frac{42}{PB} = \frac{r}{PC} = k$ (Let)



So, $k(PB - PA) = 42 - 12 = 30$ .....(1)

$\Rightarrow k(AB) = 30$

$\Rightarrow k(54) = 30 \Rightarrow \quad k = \frac{30}{54} = \frac{5}{9}$ .....(2)

Now, $k(PC - PB) = (r - 42)$

$k(BC) = (r - 42)$

$\Rightarrow \frac{5}{9}(42 + r) = (r - 42)$

$\frac{r+42}{r-42} = \frac{9}{5} \Rightarrow \frac{2r}{84} = \frac{14}{4} = \frac{7}{2} \Rightarrow r = 147$

11. If the circles

$x^2 + y^2 + 2ax + 2by + c = 0$

and $x^2 + y^2 + 2bx + 2ay + c = 0$

where $c > 0$, have exactly one point in common then the value of $\frac{(a+b)^2}{2c}$ is :

यदि वृत

$x^2 + y^2 + 2ax + 2by + c = 0$

तथा $x^2 + y^2 + 2bx + 2ay + c = 0$

जहाँ $c > 0$, का केवल एक बिन्दु उभयनिष्ठ है तो $\frac{(a+b)^2}{2c}$ का मान होगा :

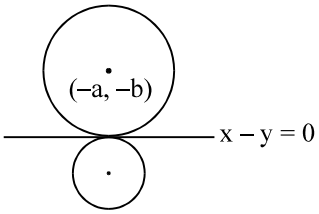
(A) 1 (B) $\sqrt{2}$ (C) 2 (D) 1/2

Ans. A

Sol. radical axis $x - y = 0$ ....(1)

centre of 1st circle is $(-a, -b)$ and $r = \sqrt{a^2 + b^2 - c}$

now perpendicular distance from centre on radical axis = radius of the circle



$(a - b)^2 = 2[a^2 + b^2 - c]$

$\Rightarrow \quad a^2 + b^2 - 2ab = 2[a^2 + b^2 - c]$

$\Rightarrow \quad a^2 + b^2 + 2ab = 2c \quad \Rightarrow \quad \dfrac{(a+b)^2}{2c} = 1$

Alternatively: $2x^2 + 2(a + b)x + c = 0$; put $D = 0$

$4(a + b)^2 = 8c \quad \Rightarrow \quad \dfrac{(a+b)^2}{2c} = 1$

12. Three concentric circles of which the biggest is $x^2 + y^2 = 1$, have their radii in A.P. If the line $y = x + 1$ cuts all the circles in real and distinct points. The interval in which the common difference of the A.P. will lie is :

तीन संकेन्द्रीय वृत में सबसे बड़ी त्रिज्या वाला वृत $x^2 + y^2 = 1$ है तथा उनकी त्रिज्याऐं समान्तर श्रेणी में है यदि रेखा $y = x + 1$ तीनों वृत्तों को वास्तविक एवं भिन्न–भिनन बिन्दुओं पर काटे तो वह अन्तराल बताओं जिसमें सर्वान्तर आयेगा :

(A) $\left(0, \dfrac{1}{4}\right)$ (B) $\left(0, \dfrac{1}{2\sqrt{2}}\right)$ (C) $\left(0, \dfrac{2-\sqrt{2}}{4}\right)$ (D) none

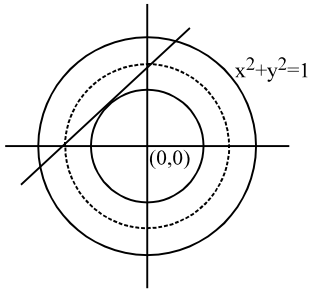
Ans. C

Sol. Radius of circle are r_1, r_2 and 1

line $y = x + 1$

perpendicular from (0, 0) on line $y = x + 1$

$= \dfrac{1}{\sqrt{2}}$



now $r_1 > \dfrac{1}{\sqrt{2}}$ but $r_1 = 1 - 2d$

hence $1 - 2d > \dfrac{1}{\sqrt{2}}$; $\dfrac{\sqrt{2}-1}{\sqrt{2}} > 2d$; $d < \dfrac{\sqrt{2}-1}{2\sqrt{2}}$

$\therefore \quad d = \dfrac{\sqrt{2}-1}{2\sqrt{2}}$

Aliter : Equation of circle are

$x^2 + y^2 = 1$

$x^2 + y^2 = (1 - d)^2$

$x^2 + y^2 = (1 - 2d)^2$

$\Rightarrow$ solve any of circle with line $y = x + 1$

e.g. $x^2 + y^2 = (1 - d)^2 \Rightarrow 2x^2 + 2x + 2d - d^2 = 0$ cuts the circle in real and distinct point hence $\Delta > 0$

$\Rightarrow \quad 2d^2 - 4d + 1 > 0 \quad \Rightarrow \quad d = \dfrac{2 \pm \sqrt{2}}{4}$

13. A tangent at a point on the circle $x^2 + y^2 = a^2$ intersects a concentric circle C at two points P and Q. The tangents to the circle C at P and Q meet at a point on the circle $x^2 + y^2 = b^2$ then the equation of circle 'C' is

वृत $x^2 + y^2 = a^2$ के किसी बिन्दु पर खीची गयी स्पर्श रेखा एक दूसरी संकेन्द्रीय वृत C को P एवं Q पर मिलती है। वृत C पर बिन्दु P एवं Q से खीची गयी स्पर्श रेखा यदि वृत $x^2 + y^2 = b^2$ पर मिलती है तो वृत 'C' का समीकरण होगा–

(A) $x^2 + y^2 = ab$ (B) $x^2 + y^2 = (a - b)^2$ (C) $x^2 + y^2 = (a + b)^2$ (D) $x^2 + y^2 = a^2 + b^2$

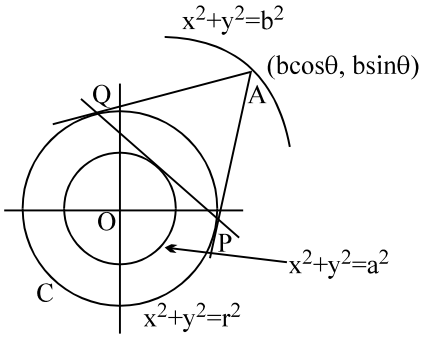
Ans. A

Sol. C.O.C. of the point A w.r.t. $x^2 + y^2 = r^2$ is

$xb\cos\theta + yb\sin\theta = r^2$ .....(1)

This must be a tangent to the circle $x^2 + y^2 = a^2$

$$\left|\frac{r^2}{\sqrt{b^2\cos^2\theta + b^2\sin^2\theta}}\right| = a \Rightarrow r^2 = ab$$



14. Tangents are drawn to the circle $x^2 + y^2 = 1$ at the points where it is met by the circles, $x^2 + y^2 - (\lambda + 6)x + (8 - 2\lambda)y - 3 = 0$. λ being the variable. The locus of the point of intersection of these tangents is:

$C_1 : x^2 + y^2 = 1$

$C_2 : x^2 + y^2 - (\lambda + 6)x + (8 - 2\lambda)y - 3 = 0.$

C_1 एवं C_2 के उभयनिष्ठ बिन्दु से C_1 पर खीची गयी स्पर्श रेखा के प्रतिच्छेद बिन्दु का बिन्दुपथ बताओं जबकि λ चर हो :

(A) $2x - y + 10 = 0$ (B) $x + 2y - 10 = 0$ (C) $x - 2y + 10 = 0$ (D) $2x + y - 10 = 0$

Ans. A

Sol. Locus of point of intersection of tangents

chord of contact of (x_1, y_1) w.r.t. $x^2 + y^2 = 1$ is $xx_1 + yy_1 = 1$ .....(1)

AB is also common chord between two circles

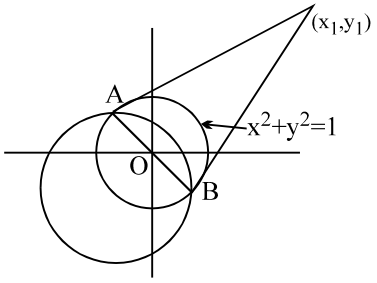
$\therefore \quad -1 + (\lambda + 6)x - (8 - 2\lambda)y + 3 = 0$

$\Rightarrow \quad (\lambda + 6)x - (8 - 2\lambda)y + 2 = 0$ .....(2)

comparing (1) and (2) we get

$$\frac{x_1}{\lambda + 6} = \frac{y}{2\lambda - 8} = \frac{-1}{2}$$

eliminate $\lambda \quad \Rightarrow \quad 2x - y + 10 = 0$



Paragraph for Question No. 15 to 17

Let each of the circles
$S_1 \equiv x^2 + y^2 + 4y - 1 = 0$, $S_2 \equiv x^2 + y^2 + 6x + y + 8 = 0$, $S_3 \equiv x^2 + y^2 - 4x - 4y - 37 = 0$
touches the other two. Let P_1, P_2, P_3 be the points of contact of S_1 and S_2, S_2 and S_3, S_3 and S_1 respectively and C_1, C_2, C_3 be the centres of S_1, S_2, S_3 respectively.

माना $S_1 \equiv x^2 + y^2 + 4y - 1 = 0$, $S_2 \equiv x^2 + y^2 + 6x + y + 8 = 0$, $S_3 \equiv x^2 + y^2 - 4x - 4y - 37 = 0$
एक दूसरे को स्पर्श कर रहे है यदि P_1, P_2 एवं P_3 क्रमशः S_1 एवं S_2, S_2 एवं S_3, S_3 एवं S_1 का स्पर्श बिन्दु है तथा वृत्तों S_1, S_2 तथा S_3 के केन्द्र C_1, C_2 तथा C_3 हैं, तो

15. The coordinates of P_1 are :

P_1 का निर्देशांक :

(A) $(2, -1)$ (B) $(2, 1)$ (C) $(-2, 1)$ (D) $(-2, -1)$

Ans. D

Sol.

16. The ratio $\frac{\text{area}(\Delta P_1P_2P_3)}{\text{area}(\Delta C_1C_2C_3)}$ is equal to :

$\frac{(\Delta P_1P_2P_3) \text{ का क्षेत्रफल}}{(\Delta C_1C_2C_3) \text{का क्षेत्रफल}}$ का मान होगा :

(A) 3 : 2 (B) 2 : 5 (C) 5 : 3 (D) 3 : 2

Ans. B

Sol.

17. P_2 and P_3 are image of each other with respect to line :

P_2 एवं P_3 किस रेखा के सापेक्ष एक दूसरे के प्रतिबिम्ब है :

(A) $y = x + 1$ (B) $y = -x$ (C) $y = x$ (D) $y = -x + 2$

Ans. C

Sol. S_1 has centre $C_1(0, -2)$ and radius $r_1 = \sqrt{5}$.

S_2 has centre $C_2 \left(-3, \frac{-1}{2}\right)$ and radius $r_2 = \frac{\sqrt{5}}{2}$.

S_3 has centre $C_3(2, 2)$ and radius $r_3 = 3\sqrt{5}$.

Also, $C_1C_2 = \sqrt{9+\frac{9}{4}} = \frac{3\sqrt{5}}{2} = r_1 + r_2 \Rightarrow S_1$ and S_2 touch each other externally. .....(1)

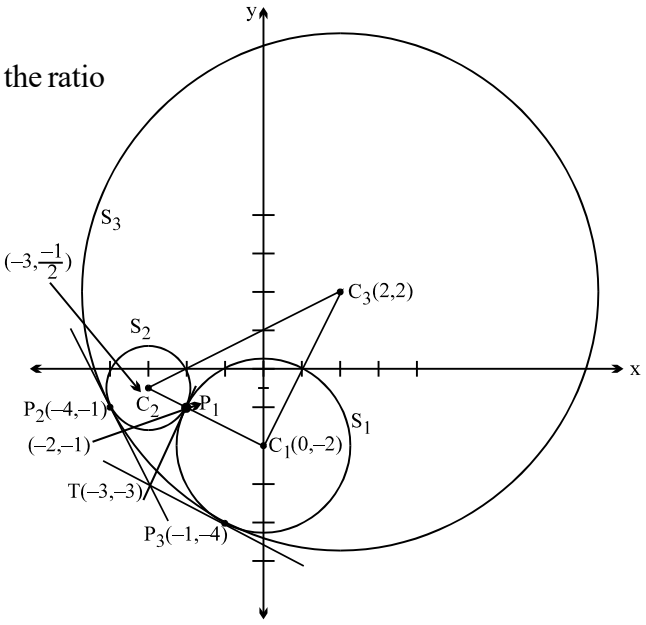
Also, $C_2C_3 = \sqrt{25+\frac{25}{4}} = \frac{5\sqrt{5}}{2} = r_3 - r_2 \Rightarrow S_2$ and S_3 touch each other internally. .....(2)

And $C_1C_3 = \sqrt{4+16} = 2\sqrt{5} = r_3 - r_1 \Rightarrow S_1$ and S_3 touch each other internally. .....(3)

(A) (i) The point of contact P_1 divides C_1C_2 in the ratio $r_1 : r_2 = 2 : 1 \Rightarrow P_1 \equiv (-2, -1)$ Ans.(i).

(ii) The point of contact P_2 divides C_2C_3 in the ratio $-r_2 : r_3 = -1 : 6 \Rightarrow P_2 \equiv (-4, -1)$.

(iii) The point of contact P_3 divides C_3C_1 in the ratio $r_3 : -r_1 = 3 : -1 \Rightarrow P_3 \equiv (-1, -4)$.



(B) Area $(\Delta P_1P_2P_3) = \frac{1}{2}\left|\begin{vmatrix} -2 & -1 & 1 \\ -4 & -1 & 1 \\ -1 & -4 & 1 \end{vmatrix}\right| = 3$

and area $(\Delta C_1C_2C_3) = \frac{1}{2}\left|\begin{vmatrix} 0 & -2 & 1 \\ -3 & \frac{-1}{2} & 1 \\ 2 & 2 & 1 \end{vmatrix}\right| = \frac{15}{2}$

$$\therefore \frac{\text{area}(\Delta P_1P_2P_3)}{\text{area}(\Delta C_1C_2C_3)} = \frac{3}{\frac{15}{2}} = 2 : 5$$

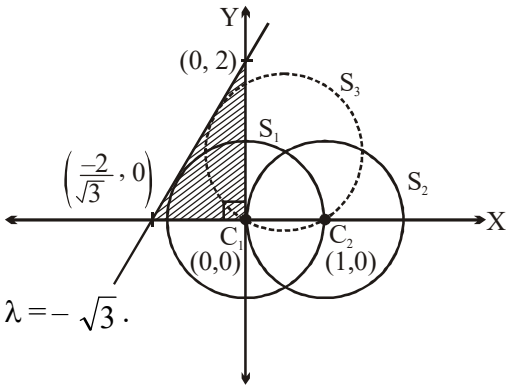
(C) As P_2 is $(-4, -1)$ and P_3 $(-1, -4)$, so P_2 and P_3 are image of each other with respect to line $y = x$ Ans.

18. Let S_1 and S_2 be circles of unit radius with centres at $C_1(0, 0)$ and $C_2(1, 0)$ respectively. S_3 is a circle of unit radius passing through C_1 and C_2 with its centre C_3 lying above x-axis. Find y-intercept of the common tangent to S_1 and S_3 which does not intersect S_2.

इकाई त्रिज्या के दो वृत S_1 एवं S_2 का केन्द्र $C_1(0, 0)$ तथा $C_2(1, 0)$ है इकाई त्रिज्या का वृत S_3, C_1 एवं C_2 से जा रहा है तथा इसका केन्द्र C_3 x-अक्ष के उपर स्थित है तो S_1 एवं S_3 के उस उभयनिष्ठ स्पर्श रेखा का y-अक्ष पर कटा अन्तःखण्ड की लम्बाई बताओं जो कि S_2 का नही काटता।

Ans. 2

Sol. Equation of any circle passing through
$C_1(0, 0)$ and $C_2(1,0)$ is
$(x - 1)(x - 0) + (y - 0)(y - 0) + \lambda y = 0$
$\Rightarrow \quad x^2 + y^2 - x + \lambda y = 0$



If it represents S_3, then $1 = \frac{1}{4} + \frac{\lambda^2}{4} \Rightarrow \lambda = \pm\sqrt{3}$

As the centre C_3 of circle S_3, lies above the x-axis, so we take $\lambda = -\sqrt{3}$.

$\therefore \quad$ Equation of circle S_3 is $x^2 + y^2 - x - \sqrt{3}\, y = 0$

As S_1 and S_3 intersect and are of unit radius, so their common tangents are parallel to the line joining their centres

C_1 (0, 0) and $C_3 \equiv \left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$

So, let the equation of common tangent of S_1 and S_3 be $\sqrt{3}x - y + k = 0$

Since, it touches S_1, so $\frac{|k|}{\sqrt{3+1}} = 1 \Rightarrow k = \pm 2$

$\therefore$ From the figure, we observe that the required tangent makes positive intercept on the y-axis and negative intercept on the x-axis, so its equation is

$\sqrt{3}x - y + 2 = 0 \Rightarrow$ y-intercept $= 2$.

Special DPP on Circle **DPP # 8**

1. The acute angle between the line $3x - 4y = 5$ and the circle $x^2 + y^2 - 4x + 2y - 4 = 0$ is θ, then $\sin\theta$ equals :

   यदि रेखा $3x - 4y = 5$ तथा वृत्त $x^2 + y^2 - 4x + 2y - 4 = 0$ के बीच न्यून कोण का मान θ है, तो $\sin\theta$ होगा :

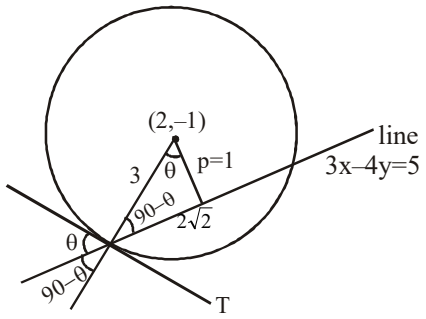
   (A) $\frac{\sqrt{5}}{3}$ (B) $\frac{1}{3}$ (C) $\frac{\sqrt{2}}{3}$ (D) $\frac{2\sqrt{2}}{3}$

Ans. D

Sol. θ is the angle between the tangent and the line
Circle with centre $(2, -1)$ and $r = 3$
perpendicular from centre on $3x - 4y = 5$ is

$$p = \left|\frac{6+4-5}{5}\right| = 1 \Rightarrow \sin(90-\theta) = \frac{1}{3} \Rightarrow \cos\theta = \frac{1}{3}$$

$$\sin\theta = \frac{2\sqrt{2}}{3}$$



2. Two congruent circles with centres at (2, 3) and (5, 6) which intersect at right angles has radius equal to :

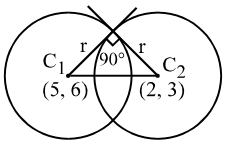
   (A) $2\sqrt{2}$ (B) 3 (C) 4 (D) none

   दो सर्वागसम वृत का केन्द्र (2, 3) तथा (5, 6) यदि एक दूसरे को लाम्बिक प्रतिच्छेदित (orthogonally intersect) करते है तो r का मान होगा :

   (A) $2\sqrt{2}$ (B) 3 (C) 4 (D) कोई नहीं

Ans. B

Sol. $2r^2 = 3^2 + 3^2 = 18$
$r^2 = 9$
$r = 3$



3. P is a point (a, b) in the first quadrant. If the two circles which pass through P and touch both the co-ordinate axes cut at right angles, then :

   P (a, b) प्रथम चतुर्थांश में है तथा दो वृत P से गुजरते है, दोनों अक्षों को स्पर्श करते है तथा 90° पर काटते है तो सही सम्बन्ध को चुनो :

   (A) $a^2 - 6ab + b^2 = 0$ (B) $a^2 + 2ab - b^2 = 0$ (C) $a^2 - 4ab + b^2 = 0$ (D) $a^2 - 8ab + b^2 = 0$

Ans. C

Sol. Equation of the two circles be $(x - r)^2 + (y - r)^2 = r^2$

i.e. $x^2 + y^2 - 2rx - 2ry + r^2 = 0$ where $r = r_1$ and r_2. Condition of orthogonality gives

$2 r_1 r_2 + 2 r_1 r_2 = r_1^2 + r_2^2 \Rightarrow 4 r_1 r_2 = r_1^2 + r_2^2$. Circle passes through (a, b)

$\Rightarrow$ $a^2 + b^2 - 2ra - 2rb + r^2 = 0$ i.e. $r^2 - 2r(a + b) + a^2 + b^2 = 0$

$r_1 + r_2 = 2(a + b)$ and $r_1 r_2 = a^2 + b^2$

4. The equation of a circle which touches the line $x + y = 5$ at N(–2, 7) and cuts the circle $x^2 + y^2 + 4x - 6y + 9 = 0$ orthogonally, is :

वृत रेखा $x + y = 5$ को N(–2, 7) पर स्पर्श काटता है तथा एक दूसरे वृत $x^2 + y^2 + 4x - 6y + 9 = 0$ को लम्बकोणीय प्रतिच्छेदित करता है तो उसका समीकरण होगा :

(A) $x^2 + y^2 + 7x - 11y + 38 = 0$ (B) $x^2 + y^2 = 53$

(C) $x^2 + y^2 + x - y - 44 = 0$ (D) $x^2 + y^2 - x + y - 62 = 0$

Ans. A

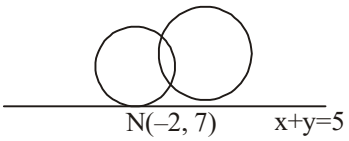
Sol. Equation of required circle is

$(x + 2)^2 + (y - 7)^2 + \lambda(x + y - 5) = 0$

$\Rightarrow$ $x^2 + y^2 + (\lambda + 4)x + (\lambda - 14)y + (53 - 5\lambda) = 0$

Now, by condition of orthogonality, we get

$$2\left[(2)\left(\frac{\lambda+4}{2}\right) + (-3)\left(\frac{\lambda-14}{3}\right)\right] = 9 + 53 - 5\lambda$$



$\Rightarrow$ $4\lambda = 12$

$\therefore$ $\lambda = 3$

Hence the required equation of circle is $x^2 + y^2 + 7x - 11y + 38 = 0$ Ans.

5. Equation of the circle which cuts the circle $x^2 + y^2 + 2x + 4y - 4 = 0$ and the lines $xy - 2x - y + 2 = 0$ orthogonally, is :

(A) $x^2 + y^2 - 2x - 4y - 6 = 0$ (B) $x^2 + y^2 - 2x - 4y + 6 = 0$

(C) $x^2 + y^2 - 2x - 4y + 12 = 0$ (D) Not possible to determine.

उस वृत का समीकरण जो कि वृत $x^2 + y^2 + 2x + 4y - 4 = 0$ तथा रेखा युग्म $xy - 2x - y + 2 = 0$ को लाम्बिक प्रतिच्छेदित करता है, क्या होगा :

(A) $x^2 + y^2 - 2x - 4y - 6 = 0$ (B) $x^2 + y^2 - 2x - 4y + 6 = 0$

(C) $x^2 + y^2 - 2x - 4y + 12 = 0$ (D) ज्ञात करना सम्भव नहीं है

Ans. A

Sol. Line are $(x - 1)(y - 2) = 0 \Rightarrow x = 1$ and $y = 2$

Centre (1, 2) i.e. $-g = 1$ and $-f = 2$

Let the equation of the required circle is $x^2 + y^2 + 2gx + 2fy + c = 0$ ....(1)

As (1) is orthogonal to $x^2 + y^2 + 2x + 4y - 4 = 0$

$\therefore \quad -2-8=-4+c \quad \Rightarrow \quad c=-6$

Hence $x^2+y^2-2x-4y-6=0$ Ans.

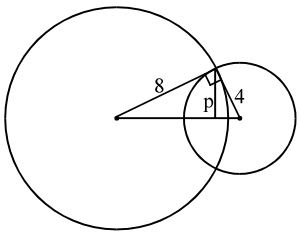
6. Two circles whose radii are equal to 4 and 8 intersect at right angles. The length of their common chord is :

दो वृत की त्रिज्या 4 एवं 8 है तथा वे एक दूसरे को 90º पर प्रतिच्छेदित करते है तो उनके उभयनिष्ठ जीवा की लम्बाई होगी :

(A) $\frac{16}{\sqrt{5}}$ (B) 8 (C) $4\sqrt{6}$ (D) $\frac{8\sqrt{5}}{5}$

Ans. A

Sol. Distance 'd' between the centres is = $\sqrt{8^2+4^2}=4\sqrt{5}$



Also $4\sqrt{5}\ .\ p=8.4 \Rightarrow p=\frac{8}{\sqrt{5}}$

$\Rightarrow$ length of common chord is $\frac{16}{\sqrt{5}}$

7. If a circle passes through the point (a , b) and cuts the circle $x^2+y^2=K^2$ orthogonally, then the equation of the locus of itscentre is :

यदि वृत (a , b) से जाता है तथा वृत $x^2+y^2=K^2$ को लम्बकोणीय प्रतिच्छेदित करता है तो वृत केन्द्र का बिन्दुपथ होगा :

(A) $2ax+2by-(a^2+b^2+K^2)=0$ (B) $2ax+2by-(a^2-b^2+K^2)=0$

(C) $x^2+y^2-3ax-4by+(a^2+b^2-K^2)=0$ (D) $x^2+y^2-2ax-3by+(a^2-b^2-K^2)=0$

Ans. A

Sol. Let $S=x^2+y^2+2gx+2fy+c=0$; where $c=k^2$

8. A variable circle cuts each of the circles $x^2+y^2-2x=0$ and $x^2+y^2-4x-5=0$ orthogonally. The variable circle passes through two fixed points whose co–ordinates are :

वृतो $x^2+y^2-2x=0$ तथा $x^2+y^2-4x-5=0$ को लाम्बिक काटने वाली वृत चर वृत दो स्थिर बिन्दुओं से होकर जायेगी जिसका निर्देशांक होगा :

(A) $\left(\frac{-5\pm\sqrt{3}}{2},0\right)$ (B) $\left(\frac{-5\pm3\sqrt{5}}{2},0\right)$ (C) $\left(\frac{-5\pm5\sqrt{3}}{2},0\right)$ (D) $\left(\frac{-5\pm\sqrt{5}}{2},0\right)$

Ans. B

Sol. $c=-5$; $g=5/2 \Rightarrow$ B

9. The angle at which the circles $(x-1)^2+y^2=10$ and $x^2+(y-2)^2=5$ intersect is :

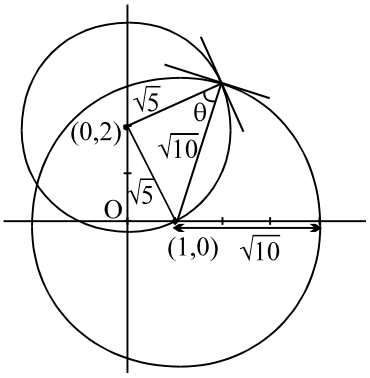
वृत $(x-1)^2 + y^2 = 10$ तथा $x^2 + (y-2)^2 = 5$ किस कोण पर प्रतिच्छेदित करेगें :

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{2}$

Ans. B

Sol. $\cos\theta = \frac{10+5-5}{2\cdot\sqrt{10}\cdot\sqrt{5}} = \frac{1}{\sqrt{2}}$

$\Rightarrow \quad \theta = \frac{\pi}{4}$



Paragraph for Question Nos. 10 to 13

Consider the circle S: $x^2 + y^2 - 4x - 1 = 0$ and the line L : $y = 3x - 1$. If the line L cuts the circle at A and B then

यदि S: $x^2 + y^2 - 4x - 1 = 0$ तथा L : $y = 3x - 1$. यदि रेखा वृत को बिन्दु A एवं B पर काटती है तो

10. Length of the chord AB equal :

जीवा AB की लम्बाई होगी :

(A) $2\sqrt{5}$ (B) $\sqrt{5}$ (C) $5\sqrt{2}$ (D) $\sqrt{10}$

Ans. D

Sol.

11. The angle subtended by the chord AB in the minor arc of S is :

जीवा AB द्वारा लघु वृत चाप S पर आन्तरित कोण होगा :

(A) $\frac{3\pi}{4}$ (B) $\frac{5\pi}{6}$ (C) $\frac{2\pi}{3}$ (D) $\frac{\pi}{4}$

Ans. A

Sol.

12. Acute angle between the line L and the circle S is :

रेखा L एवं वृत S के बीच न्यूनकोण होगा :

(A) $\frac{\pi}{2}$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi}{4}$ (D) $\frac{\pi}{6}$

Ans. C

Sol.

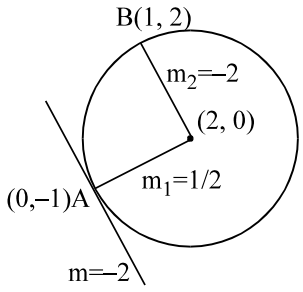
13. If the equation of the circle on AB as diameter is of the form $x^2 + y^2 + ax + by + c = 0$ then the magnitude of the vector $\vec{V} = a\hat{i} + b\hat{j} + c\hat{k}$ has the value equal to :

AB को व्यास मानकर खीचें वृत का समीकरण यदि $x^2 + y^2 + ax + by + c = 0$ है तो सदिश $\vec{V} = a\hat{i} + b\hat{j} + c\hat{k}$ का परिमाण होगा:

(A) $\sqrt{8}$ (B) $\sqrt{6}$ (C) $\sqrt{9}$ (D) $\sqrt{10}$

Ans. B

Sol. Solve S and L and interpret from the figure.



Paragraph for question nos. 14 to 16

Consider the family of circles $x^2 + y^2 - 2x - 2\lambda y - 8 = 0$ passing through two fixed points A and B. Also S = 0 is a circle of this family, the tangent to which at A and B intersect on the line $x + 2y + 5 = 0$.

वृतो का परिवार $x^2 + y^2 - 2x - 2\lambda y - 8 = 0$ यदि दो स्थिर बिन्दु A एवं B से गुजरता है। S = 0 इस परिवार का सदस्य है जिस पर बिन्दु A एवं B से खीचीं स्पर्श रेखा $x + 2y + 5 = 0$ पर मिलती है।

14. The distance between the points A and B, is :

A एवं B के बीच की दूरी :

(A) 4 (B) $4\sqrt{2}$ (C) 6 (D) 8

Ans. C

Sol. We have $(x^2 + y^2 - 2x - 8) - 2\lambda y = 0$, which is the form of $S + \lambda L = 0$

$\therefore$ On solving $S = 0$ and $L = 0$ where $S \equiv x^2 + y^2 - 2x - 8 = 0$ and $L \equiv y = 0$

Put $y = 0 \Rightarrow x^2 - 2x - 8 = 0$ $(x - 4)(x + 2) = 0$ $x = -2$ or 4

$\therefore$ $A \equiv (4, 0)$; $B \equiv (-2, 0)$ $\Rightarrow$ Distance between A and B $= 6$ Ans.

15. The area of an equilateral triangle inscribed in S, is :

वृत S में बनाये गये समबाहु त्रिभुज का क्षेत्रफल होगा :

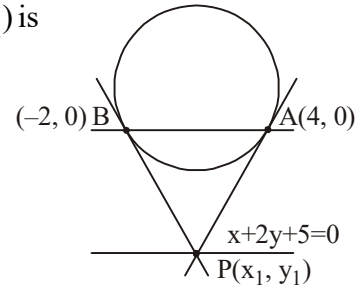
(A) $\frac{27\sqrt{3}}{4}$ (B) $\frac{9\sqrt{3}}{2}$ (C) $\frac{27\sqrt{3}}{2}$ (D) $9\sqrt{3}$

Ans. C

Sol. Equation of AB which is chord of contact with respect to $P(x_1, y_1)$ is

$$xx_1 + yy_1 - 1(x + x_1) - \lambda(y + y_1) - 8 = 0$$

$\Rightarrow$ $x(x_1 - 1) + y(y_1 - \lambda) - (x_1 + \lambda y_1 + 8) = 0$ ....(1)



Also equation of AB is x-axis i.e

$$0 \cdot x + 1 \cdot y + 0 \cdot c = 0$$ ....(2)

$\therefore$ On comparing (1) & (2), we get

$$\frac{x_1 - 1}{0} = \frac{y_1 - \lambda}{1} = \frac{x_1 + \lambda y_1 + 8}{0} = k \text{ say}$$

$\Rightarrow$ $x_1 - 1 = 0$ $\Rightarrow$ $x_1 = 1$; $y_1 = \lambda + k$

and $x_1 + \lambda y_1 + 8 = 0$ $\Rightarrow$ $9 + \lambda y_1 = 0$ $\Rightarrow$ $y_1 = -\frac{9}{\lambda}$

Also $x_1 + 2y_1 + 5 = 0 \Rightarrow 1 - \frac{18}{\lambda} + 5 = 0$ $\Rightarrow$ $\lambda = \frac{18}{6}$ $\Rightarrow$ $\lambda = 3$

$\therefore$ Required equation of circle $S = 0$ is $x^2 + y^2 - 2x - 6y - 8 = 0$

Now radius of circle $S = 0$ is $r = \sqrt{1 + 9 + 8} = 3\sqrt{2}$

Hence area of equilateral triangle inscribed in a circle of radius 'r' $= \frac{3\sqrt{3}}{4} r^2 = \frac{3\sqrt{3}}{4} \cdot 18 = \frac{27\sqrt{3}}{2}$ Ans.

16. If the circle $x^2 + y^2 - 10x + 2y + c = 0$ is orthogonal to $S = 0$, then the value of c equals :

यदि वृत $x^2 + y^2 - 10x + 2y + c = 0$, वृत $S = 0$ के लाम्बिक हो तो c का मान होगा :

(A) 8 (B) 9 (C) 10 (D) 12

Ans. D

Sol. If the circle $x^2 + y^2 - 10x + 2y + c = 0$ is orthogonal to $S \equiv x^2 + y^2 - 2x - 6y - 8 = 0$,
so by applying condition of orthogonality, we get $2\left[(-5)(-1) + (1)(-3)\right] = c - 8 \Rightarrow 4 = c - 8$
Hence $c = 12$ Ans.

17. Find the radical centre of the following set of circles
$x^2 + y^2 - 3x - 6y + 14 = 0$; $x^2 + y^2 - x - 4y + 8 = 0$; $x^2 + y^2 + 2x - 6y + 9 = 0$

वृत $x^2 + y^2 - 3x - 6y + 14 = 0$; $x^2 + y^2 - x - 4y + 8 = 0$ तथा $x^2 + y^2 + 2x - 6y + 9 = 0$ का मूलाक्ष केन्द्र (radical centre) क्या होगा।

Ans. (1, 2)

Sol.

18. Find the equation to the circle orthogonal to the two circles $x^2 + y^2 - 4x - 6y + 11 = 0$; $x^2 + y^2 - 10x - 4y + 21 = 0$ and has $2x + 3y = 7$ as diameter.

उस वृत का समीकरण बताओं जो कि दो वृतो $x^2 + y^2 - 4x - 6y + 11 = 0$ तथा $x^2 + y^2 - 10x - 4y + 21 = 0$ को लम्बकोणीय काटते है तथा जिसका व्यास $2x + 3y = 7$ है।

Ans. $x^2 + y^2 - 4x - 2y + 3 = 0$

Sol. Let the equation of t he required circle be

$$x^2 + y^2 + 2gx + 2fy + c = 0 \quad ...(1)$$

Now the circle (1) cut the given circles orthogonally, we have

$$2g(-2) + 2f(-3) = c + 11 \Rightarrow -4g - 6f - c = 1 \quad ...(2)$$

Similarly $-10g - 4f - c = 21$ ...(3)

Further the line $2x + 3y = 7$ is the diameter of (1), therefore the $(-g, -f)$ of (1) lies on it so that

$$-2g - 3f = 7 \quad ...(4)$$

From (2) and (3) by subtraction, we have

$$3g - f = -5 \quad ...(5)$$

Solving (4) and (5), we get

$$g = -2 \text{ and } f = -1$$

Putting the values of f and g in (2), we get

$$-10(-2) - 4(-1) - c = 21 \Rightarrow c = 3$$

Hence the equation of the required circle is

$$x^2 + y^2 - 4x - 2y + 3 = 0 \text{ Ans.}$$

19. Find the equation to the circle, cutting orthogonally each of the following circles $x^2 + y^2 - 2x + 3y - 7 = 0$; $x^2 + y^2 + 5x - 5y + 9 = 0$; $x^2 + y^2 + 7x - 9y + 29 = 0$.

उस वृत का समीकरण बताओ जो वृतों $x^2 + y^2 - 2x + 3y - 7 = 0$; $x^2 + y^2 + 5x - 5y + 9 = 0$; $x^2 + y^2 + 7x - 9y + 29 = 0$

को लाम्बिक प्रतिच्छेद करता है।

Ans. $x^2 + y^2 - 16x - 18y - 4 = 0$

Sol.

20. Tangents drawn from P(1, 8) to the circle $x^2 + y^2 - 6x - 4y - 11 = 0$ touches the circle at the points A and B respectively. If the radius of the circle which passes through the points of intersection of circles $x^2 + y^2 - 2x - 6y + 6 = 0$ and $x^2 + y^2 + 2x - 6y + 6 = 0$ and intersects the circumcircle of the ΔPAB orthogonally is equal to $\frac{\sqrt{p}}{q}$ where $p, q \in N$, then find the minimum value of $(p + q)$.

P(1, 8) से वृत $x^2 + y^2 - 6x - 4y - 11 = 0$ पर खीची गयी स्पर्श रेखा PA एवं PB है। वृत $x^2 + y^2 - 2x - 6y + 6 = 0$ तथा $x^2 + y^2 + 2x - 6y + 6 = 0$ के प्रतिच्छेद बिन्दु से जाने वाला तथा ΔPAB के परिवृत को लम्बकोणीय प्रतिच्छेदित करने वाले वृत की त्रिज्या यदि $\frac{\sqrt{p}}{q}$ है जहाँ $p, q \in N$ तो $(p + q)$ का न्यूनतम मान बताओ।

Ans. 77

Sol. Equation of circle circumscribing ΔPAB is

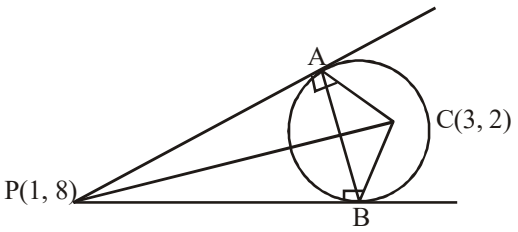
$(x - 1)(x - 3) + (y - 8)(y - 2) = 0 \quad \Rightarrow \quad x^2 + y^2 - 4x - 10y + 19 = 0$ .....(i)

Equation of circle passing through points of intersection of circles $x^2 + y^2 - 2x - 6y + 6 = 0$ and $x^2 + y^2 + 2x - 6y + 6 = 0$ is given by $(x^2 + y^2 - 2x - 6y + 6) + \lambda(x^2 + y^2 + 2x - 6y + 6) = 0$

$\Rightarrow \quad x^2 + y^2 + \frac{(2\lambda - 2)}{\lambda + 1}x - 6y + 6 = 0$ .....(ii)

As circle (ii) is orthogonal to circle (i), so

$-2\left(\frac{2\lambda - 2}{\lambda + 1}\right) - 5(-6) = 19 + 6$



$\Rightarrow \quad 2\left(\frac{2\lambda - 2}{\lambda + 1}\right) = 5 \Rightarrow 4\lambda - 4 = 5\lambda + 5 \Rightarrow \lambda = -9$

Hence required equation of circle is $x^2 + y^2 + \frac{5}{2}x - 6y + 6 = 0$

Radius $= \sqrt{\frac{25}{16} + 9 - 6} = \frac{\sqrt{73}}{4} = \frac{\sqrt{p}}{q}$

$\therefore p = 73, q = 4$

Hence minimum value of $(p + q) = 73 + 4 = 77$ Ans.